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UNIT-2:

INNER PRODUCT SPACES

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Defn:

Suppose u, v and w are all vectors in a vector space V and k is any scalar, then an inner product on the real vector space V is a function that associates a real no. with each pair of vectors u and v , a real no. denoted by $\langle u, v \rangle$ that satisfies the following axioms,

- 1) $\langle u, v \rangle = \langle v, u \rangle$ (Symmetry axiom)
- 2) $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ (Additive axiom)
- 3) $\langle ku, v \rangle = k \langle u, v \rangle$ (Homogeneity axiom)
- 4) $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ iff $u=0$ (Positivity axiom).

A real vector space with an innerproduct is called a inner product space.

Ex-1: Euclidean Inner product on $\mathbb{R}^n$:

For $u = (u_1, u_2, \dots, u_n)$, $v = (v_1, v_2, \dots, v_n)$ in $\mathbb{R}^n$, define $\langle u, v \rangle = u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$.

(Verify) $\langle u, v \rangle$ is an innerproduct on $\mathbb{R}^n$.

Ex-2: Let $u = (u_1, u_2)$, $v = (v_1, v_2)$ in $\mathbb{R}^2$. Verify $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$ is inner product space.

Soln:

i) $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2 = 3v_1 u_1 + 2v_2 u_2 = \langle v, u \rangle$.

ii) $\langle u+v, w \rangle = 3(u_1+v_1)w_1 + 2(u_2+v_2)w_2$

$= 3u_1 w_1 + 3v_1 w_1 + 2u_2 w_2 + 2v_2 w_2$

$$3) \langle ku, v \rangle = 3k u_1 v_1 + 2k u_2 v_2 \\ = k [3u_1 v_1 + 2u_2 v_2] = k \langle u, v \rangle$$

$$4) \langle u, u \rangle = 3u_1^2 + 2u_2^2 \geq 0.$$

$$\& \langle u, u \rangle = 0 \text{ iff } u = 0.$$

$\therefore (R^2, \langle \rangle)$ is a weighted inner product space.

Note:

For u, v in R^n , $\langle u, v \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n$ is also an inner product in R^n if $w_1, w_2, \dots, w_n$ are positive real nos. which are called as weights, then $(R^n, \langle \rangle)$ is weighted innerproduct space with weights $w_1, w_2, \dots, w_n$.

Exercise - 6.1:

a) Let $\langle u, v \rangle$ be the Euclidean inner product on R^2 , and let $u = (3, -2)$, $v = (4, +5)$, $w = (-1, 6)$, and $k = -4$. Verify that.

$$a) \langle u, v \rangle = \langle v, u \rangle \Rightarrow \langle u, v \rangle = u_1 v_1 + u_2 v_2 = 3 \times 4 + (-2)(+5) \\ = 12 - 10 = 2$$

$$b) \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \quad \langle u, u \rangle = \underline{2}.$$

$$\Rightarrow \langle u+v, w \rangle = (u_1+v_1)w_1 + (u_2+v_2)w_2 \\ = (3+4)(-1) + (-2+5)(6) = -7 + 18 = 11$$

$$\langle u, w \rangle = 3(-1) + (-2)(6) = -3 - 12 = -15$$

$$\langle v, w \rangle = 4(-1) + (+5)6 = -4 + 30 = 26$$

$$\langle u, w \rangle + \langle v, w \rangle = \underline{11}$$

$$c) \langle u, u \rangle$$

$$\Rightarrow \langle u, u \rangle$$

$$\langle u, u \rangle$$

$$\langle u, u \rangle$$

$$d) \langle ku, u \rangle$$

$$\Rightarrow \langle$$

$$k$$

$$e) \langle 0, 1 \rangle$$

$$f) \text{ Repeat } \langle u, v \rangle$$

$$g) \langle u, v \rangle$$

$$\text{how } \langle u, v \rangle$$

$$\langle u, v \rangle$$

$$h) \langle u+v, u \rangle$$

$$\langle u+v, u \rangle$$

$$i) \langle u, u \rangle$$

$$\Rightarrow \langle u, u \rangle$$

$$c) \langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$$

$$\Rightarrow \langle u, v+w \rangle = (3)[4-1] + (-2)[+5+6] = 9 - 22 = -13$$

$$\langle u, v \rangle = u_1 v_1 + u_2 v_2 = (3)(4) + (-2)(+5) = 12 - 10 = 2$$

$$\langle u, w \rangle = (3)(-1) + (-2)(6) = -3 - 12 = -15$$

$$d) \langle ku, v \rangle = k \langle u, v \rangle$$

$$\Rightarrow \langle ku, v \rangle = \langle (-12, 8), (4, +5) \rangle = (-12)(4) + 8(-5) = -48 - 40 = -88$$

$$k \langle u, v \rangle = (-4)(2) = -8$$

$$e) \langle 0, v \rangle = \langle v, 0 \rangle = 0$$

f) Repeat ① for the weighted Euclidean inner product
 $\langle u, v \rangle = 4u_1 v_1 + 5u_2 v_2$

$$a) \langle u, v \rangle = \langle v, u \rangle$$

$$\langle u, v \rangle = 4u_1 v_1 + 5u_2 v_2 = 4(3)(4) + 5(-2)(5) = 48 - 50 = -2$$

$$\langle v, u \rangle = -2$$

$$b) \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

$$\langle u+v, w \rangle = 4(7)(-1) + 5(3)(6) = -28 + 90 = 62$$

$$\langle u, w \rangle + \langle v, w \rangle = 4 \times 3 \times -1 + 5 \times -2 \times 6 + 4 \times 4 \times -1 + 5 \times 5 \times 6 = -12 - 60 - 16 + 150 = 62$$

$$c) \langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$$

$$\Rightarrow \langle u, v+w \rangle = 4(3)(3) + 5(-2)(11) = 36 - 110 = -74$$

u, v

duct space

$u_1, \dots, u_n$

$u_2, \dots, u_n$ are weights, then
 x with

uct on $\mathbb{R}^2$

1, 6), and

$$3 \times 4 + (-2)(+5)$$

$$12 - 10 = 2$$

$$+18$$

$$26$$

$$-15$$

$$11$$

$$d) \langle Ku, v \rangle = K \langle u, v \rangle$$

$$e) \langle 0, v \rangle = \langle u, 0 \rangle = 0$$

8

Length and Distance in Inner product spaces:

Defn:

If V is an inner product space, then the norm (or length) of a vector u in V is denoted by $\|u\|$ and is defined by $\|u\| = \langle u, u \rangle^{1/2}$.

The distance b/w two points (vectors) u & v is denoted by $d(u, v)$ and is defined by

$$d(u, v) = \|u - v\| = \langle u - v, u - v \rangle^{1/2}$$

Example-3: Norm and distance in $\mathbb{R}^n$.

If $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ are vectors in $\mathbb{R}^n$ with the Euclidean inner product, then

$$\|u\| = \langle u, u \rangle^{1/2} = (u \cdot u)^{1/2} = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$$

$$\text{and } d(u, v) = \|u - v\| = \langle u - v, u - v \rangle^{1/2} = [(u - v) \cdot (u - v)]^{1/2} \\ = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

Note:

Norm and ~~inner product~~ distance depend on the inner product being used. If the inner product is changed, then the norms and distances b/w vectors also change.

Example-4:

For the vectors $u = (1, 0)$ and $v = (0, 1)$ in $\mathbb{R}^2$ with the Euclidean inner product, we have

$$\|u\| = \sqrt{1^2 + 0^2} = 1 \quad \text{and}$$

$$d(u, v) = \|u - v\| = \|(1, -1)\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

However, Euclidean

$$\langle u, v \rangle$$

$$\|u\| = \langle u, u \rangle^{1/2}$$

$$d(u, v) =$$

Ex: 6.1

a) Let $u =$ we inner product

$$b) \langle u, v \rangle$$

c) Let $u =$ following that do

$$d) \langle u, v \rangle$$

norm-1:

$$\langle u, v \rangle$$

$$\langle u, v \rangle$$

norm-2:

L.H.S

$$(u_1 + v_1)u$$

norm-3:

$$\langle Ku, v \rangle$$

norm-4:

$$\langle u, u \rangle$$

But

$\therefore$ An

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$\langle u, 0 \rangle = 0$

the space

the norm is defined by $\| \cdot \|$

$u \neq v$ is

y_2

v_n are products, the

$\overline{u_n^2} + (u_n - v_n)^2 + (v_n - v_n)^2$

the norm is changed; also check

$\mathbb{R}^2$ with

However, if we change to the weighted Euclidean inner product,

$\langle u, v \rangle = 3u_1v_1 + 2u_2v_2$, then we obtain,

$\|u\| = \langle u, u \rangle^{1/2} = [3(1)(1) + 2(0)(0)]^{1/2} = \sqrt{3}$ and

$d(u, v) = \|u - v\| = \langle (1, -1), (1, -1) \rangle^{1/2} = [3(1)(1) + 2(-1)(-1)]^{1/2} = \sqrt{5}$.

Ex: 6.1

Q8) Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$. Show that the following are inner products on $\mathbb{R}^2$ by verifying that the inner product axioms hold.

a) $\langle u, v \rangle = 3u_1v_1 + 5u_2v_2$

b) $\langle u, v \rangle = 4u_1v_1 + u_2v_1 + 4u_2v_2 + 4u_2v_2$

Q9) Let $u = (u_1, u_2, u_3)$, $v = (v_1, v_2, v_3)$. Determine which of the following are IP. For those that are not, list the axioms that do not hold.

a) $\langle u, v \rangle = u_1v_1 + u_3v_3$

Axiom-1:

$\langle u, u \rangle = u_1u_1 + u_3u_3$

L.H.S = R.H.S.

$\langle v, u \rangle = v_1u_1 + v_3u_3$

is $\langle u, v \rangle = \langle v, u \rangle$

Axiom-2: $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

L.H.S

R.H.S

$\Rightarrow (u_1+v_1)w_1 + (u_3+v_3)w_3 = u_1w_1 + u_3w_3 + v_1w_1 + v_3w_3$

$= \langle u, w \rangle + \langle v, w \rangle$

L.H.S = R.H.S

Axiom-3:

$\langle ku, v \rangle = (ku_1)v_1 + (ku_3)v_3 = k[u_1v_1 + u_3v_3] = k\langle u, v \rangle$

Axiom-4:

$\langle u, u \rangle = u_1^2 + u_3^2 > 0$

But $\langle u, u \rangle = 0$ if we have $u = (0, k, 0) \neq 0$.

$\therefore$ Axiom-4 fails.

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b) $\langle u, v \rangle = u_1^2 v_1^2 + u_2^2 v_2^2 + u_3^2 v_3^2$

Sol: Axiom-1: $\langle u, v \rangle = \langle v, u \rangle$

Axiom-2: $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

$$\begin{aligned} \langle u+v, w \rangle &= (u_1+v_1)^2 w_1^2 + (u_2+v_2)^2 w_2^2 + (u_3+v_3)^2 w_3^2 \\ &= (u_1^2 + 2u_1v_1 + v_1^2) w_1^2 + (u_2^2 + 2u_2v_2 + v_2^2) w_2^2 \\ &\quad + (u_3^2 + 2u_3v_3 + v_3^2) w_3^2 \\ &\neq \langle u, w \rangle + \langle v, w \rangle = u_1^2 w_1^2 + u_2^2 w_2^2 + u_3^2 w_3^2 \\ &\quad + (v_1^2 w_1^2 + v_2^2 w_2^2 + v_3^2 w_3^2) \end{aligned}$$

Axiom-2 fails

Axiom-3: $\langle ku, v \rangle = k^2 u_1^2 v_1^2 + k^2 u_2^2 v_2^2 + k^2 u_3^2 v_3^2$

$k \langle u, v \rangle = k [u_1^2 v_1^2 + u_2^2 v_2^2 + u_3^2 v_3^2]$

Axiom-3 fails.

Axiom-4:

$\langle u, u \rangle = u_1^4 + u_2^4 + u_3^4 > 0$

and $\langle u, u \rangle = 0$ iff $u = 0 = (0, 0, 0)$.

© $\langle u, v \rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$ (Yes) ② $\langle u, v \rangle = u_1v_1 - u_2v_2 + u_3v_3$ (Axiom-4 fails)

~~10) In each part use the given inner product.~~

~~11) In each part use the given inner product~~

~~Unit Circles~~

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Example -

a) Sketch
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b) Sketch th
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$\langle u, v \rangle =$

Sol: ① If $u =$
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② If u

If $\|u\|$

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Unit Circles and Spheres in inner product spaces:

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If V is an inner product space, then the set of points u in V that satisfy $\|u\|=1$ is called the unit sphere or sometimes the unit circle in V .
[In $\mathbb{R}^2$ and $\mathbb{R}^3$, these are the points that lie in 1 unit away from the origin].

Circle in $\mathbb{R}^2$, sphere in $\mathbb{R}^3$ for Euclidean I.P.

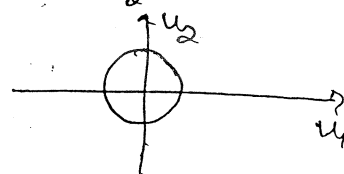
Example-5: Unusual Unit Circle in $\mathbb{R}^2$:

- Sketch the unit circle in an xy-coordinate system in $\mathbb{R}^2$ using the Euclidean inner product $\langle u, v \rangle = u_1 v_1 + u_2 v_2$.
- Sketch the unit circle in an xy-coordinate system in $\mathbb{R}^2$, using the weighted Euclidean inner product,
 $\langle u, v \rangle = \frac{1}{9} u_1 v_1 + \frac{1}{4} u_2 v_2$.

Soln:

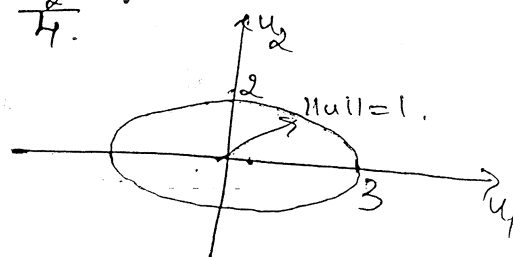
① If $u = (u_1, u_2)$, then $\|u\| = \langle u, u \rangle^{1/2} = \sqrt{u_1^2 + u_2^2}$.
Unit circle is given by, $\|u\|=1$
 $\Rightarrow u_1^2 + u_2^2 = 1$.

$\Rightarrow$ Unit circle centered at the origin



② If $u = (u_1, u_2)$, $\|u\| = \sqrt{\frac{u_1^2}{9} + \frac{u_2^2}{4}}$.

$$\begin{aligned} \text{If } \|u\|=1 &\Rightarrow \frac{u_1^2}{9} + \frac{u_2^2}{4} = 1 \\ &\Rightarrow \frac{u_1^2}{(3)^2} + \frac{u_2^2}{(2)^2} = 1 \end{aligned}$$



Unit circle is an ellipse centered at the origin.

Note:

Even though our definitions of length and distance reduce to the standard definitions when applied to $\mathbb{R}^2$ with the E.I.P., it does require a stretch of the imagination to think of the "unit circle" as a non-standard ellipse.

Kx-6.1

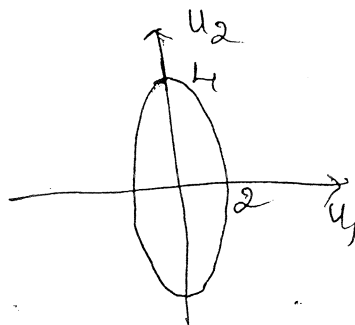
- ⑧ Sketch the unit circle in $\mathbb{R}^2$ using the given inner product.

① $\langle u, v \rangle = \frac{1}{4} u_1 v_1 + \frac{1}{16} u_2 v_2$

$\|u\| = \langle u, u \rangle^{1/2} = \sqrt{\frac{u_1^2}{4} + \frac{u_2^2}{16}}$

Unit circle is given by, $\|u\| = 1$.

$$\Rightarrow \frac{u_1^2}{(2)^2} + \frac{u_2^2}{(4)^2} = 1$$



Ellipse

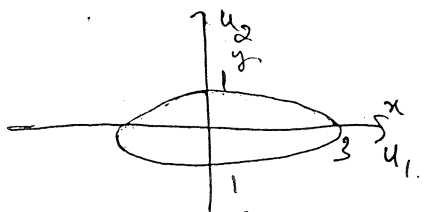
⑤ $\langle u, v \rangle = 2u_1 v_1 + u_2 v_2$

$$\|u\| = \langle u, u \rangle^{1/2} = \sqrt{2u_1^2 + u_2^2}$$

unit circle is $\|u\| = 1 \Rightarrow 2u_1^2 + u_2^2 = 1$

$$\Rightarrow \frac{u_1^2}{(1/2)^2} + \frac{u_2^2}{(1)^2} = 1$$

- ⑨ Find a weighted Euclidean I.P on $\mathbb{R}^2$ for which the unit circle is the ellipse shown in fig.



$$\frac{u_1^2}{(3)^2} + \frac{u_2^2}{(1)^2} = 1$$

$$\sqrt{\frac{u_1^2}{9} + \frac{u_2^2}{1}} = 1$$

$$\Rightarrow \|u\| = \sqrt{\frac{u_1^2}{9} + \frac{u_2^2}{1}}$$

$$\Rightarrow \langle u, v \rangle = \frac{1}{9} u_1 v_1 + u_2 v_2$$

⑩ Let $U = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$ and $V = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$

Inner product

Let $u =$

and let inner product

Alternately

$\langle u, v \rangle =$

Ex-6:

The inner

matrix

Since $\langle$

The weighted

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Note:

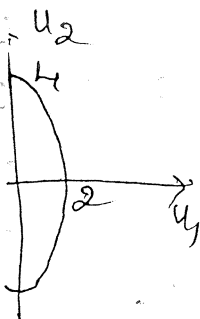
In general

$\langle u, v \rangle =$

on $\mathbb{R}^n$

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inner inner



Inner product generated by Matrices:

Let $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$, $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ be two vectors in $\mathbb{R}^n$ (expressed as $n \times 1$ matrices)

and let A be an invertible $n \times n$ matrix. Then the inner product $\langle u, v \rangle = Au \cdot Av \rightarrow \textcircled{1}$ is called inner product on $\mathbb{R}^n$ generated by A .
(dot product)

Alternatively, eqn $\textcircled{1}$ can be written as

$$\langle u, v \rangle = (Av)^T Au \text{ or } \boxed{\langle u, v \rangle = v^T A^T A u}$$

Ex-6:

The inner product on $\mathbb{R}^n$ generated by the $n \times n$ identity matrix is Euclidean inner product,

$$\text{Since } \langle u, v \rangle = Iu \cdot Iv = u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

Euclidean I.P.

The weighted Euclidean I.P $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$ is the I.P on $\mathbb{R}^2$ generated by $A = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$ because

$$\text{substituting this matrix in } \langle u, v \rangle = v^T A^T A u$$

$$= [v_1, v_2] \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$n \times 2 \times 2 \times 1$

$$= [v_1, v_2] \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 3u_1 \\ 2u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= 3u_1 v_1 + 2u_2 v_2$$

Note:

In general, the weighted Euclidean inner product,

$\langle u, v \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n$ is the inner product on $\mathbb{R}^n$ generated by

$$A = \begin{bmatrix} \sqrt{w_1} & 0 & 0 & \dots & 0 \\ 0 & \sqrt{w_2} & 0 & \dots & 0 \\ \vdots & \vdots & \sqrt{w_3} & \dots & \vdots \\ & & & \ddots & \\ 0 & 0 & 0 & \dots & \sqrt{w_n} \end{bmatrix}$$

Ex-6.1

5a) Use formula $\langle u, v \rangle = Au \cdot Av$ to show that $\langle u, v \rangle = 9u_1v_1 + 4u_2v_2$ is the inner product on $\mathbb{R}^2$ generated by $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$.

Sol: We know that $\langle u, v \rangle = Au \cdot Av$

$$= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 3u_1 \\ 2u_2 \end{bmatrix} \cdot \begin{bmatrix} 3v_1 \\ 2v_2 \end{bmatrix}$$

$$= 9u_1v_1 + 4u_2v_2$$

we know that $\langle u, v \rangle = Au \cdot Av$

$$= \begin{bmatrix} 3u_1 \\ 2u_2 \end{bmatrix} \cdot \begin{bmatrix} 3v_1 \\ 2v_2 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 9u_1 & 4u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 9u_1v_1 + 4u_2v_2$$

$u \cdot v$ can be written as $v^T u$:

$$\therefore \Rightarrow = \begin{bmatrix} 3u_1 & 2u_2 \end{bmatrix} \begin{bmatrix} 3v_1 \\ 2v_2 \end{bmatrix}$$

$$= 9u_1v_1 + 4u_2v_2$$

Thus proved.

5b) Use the inner product in a) to compute $\langle u, v \rangle$ if $u = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$.

Sol: $\langle u, v \rangle = 9u_1v_1 + 4u_2v_2 = 9(-3)(1) + 4(2)(7) = -27 + 56$
 $= 9(0)(6) + 4(-2)(2) = 54 - 24 = 29$

6a) Use formula $\langle u, v \rangle = Au \cdot Av$ to show that $\langle u, v \rangle = 5u_1v_1 - u_1v_2 - u_2v_1 + 10u_2v_2$ is the inner product on $\mathbb{R}^2$ generated by $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$

Sol:

$\langle u, v \rangle$

5b) Use the $u = \begin{pmatrix} 0 \\ - \end{pmatrix}$
 $\langle u, v \rangle$

6b) Let $u =$ given ex matrix

a) $\langle u,$

$$\therefore A = \begin{bmatrix} \end{bmatrix}$$

6c) 4(

now that
on $\mathbb{R}^2$

$$\begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\langle u, v \rangle = Au \cdot Av$$

$$= \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2u_1 + u_2 \\ -u_1 + 3u_2 \end{bmatrix} \cdot \begin{bmatrix} 2v_1 + v_2 \\ -v_1 + 3v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2u_1 + u_2 & -u_1 + 3u_2 \end{bmatrix} \begin{bmatrix} 2v_1 + v_2 \\ -v_1 + 3v_2 \end{bmatrix}$$

$$= (2u_1 + u_2)(2v_1 + v_2) + (-u_1 + 3u_2)(-v_1 + 3v_2)$$

$$= 4u_1v_1 + 2u_1v_2 + 2u_2v_1 + u_2v_2$$

$$+ u_1v_1 - 3u_1v_2 - 3u_2v_1 + 9u_2v_2$$

$$= 5u_1v_1 - u_1v_2 - u_2v_1 + 10u_2v_2$$

Thus proved.

(b) Use the inner product in part (a) to compute $\langle u, v \rangle$ if $u = (0, -3)$ and $v = (6, 2)$.

$$\langle u, v \rangle = 5u_1v_1 - u_1v_2 - u_2v_1 + 10u_2v_2$$

$$= 5(0)(6) - (0)(2) - (-3)(6) + 10(-3)(2)$$

$$= 0 - 0 + 18 - 60 = -42$$

$\langle u, v \rangle$ if

$$\begin{aligned} &= -27 \\ &\quad + 56 \\ &= 29 \end{aligned}$$

Tab
is inner

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$. In each part, the given expression is an inner product on $\mathbb{R}^2$. Find the matrix that generates it.

(a) $\langle u, v \rangle = 3u_1v_1 + 5u_2v_2$

$$w_1 = 3 \quad w_2 = 5$$

$$\therefore A = \begin{bmatrix} \sqrt{w_1} & 0 \\ 0 & \sqrt{w_2} \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

(b) $\langle u, v \rangle = 4u_1v_1 + 6u_2v_2$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{6} \end{bmatrix}$$

do (a) & (b) problems Here.

Example - 7: Inner product on M_{22} :

If $U = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$, $V = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$ are any 2×2 matrices

then $\langle U, V \rangle = \text{trace}(U^T V) = \text{trace}(V^T U) = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$ is I.P on M_{22} .
(Verify).

For Ex: If $U = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $V = \begin{bmatrix} -1 & 0 \\ 3 & 2 \end{bmatrix}$, $\langle U, V \rangle = -1 + 0 + 9 + 8 = 16$

Note:

$$\|U\| = \langle U, U \rangle^{1/2} = \sqrt{u_1^2 + u_2^2 + u_3^2 + u_4^2} \Rightarrow \|U\| = 1 \Rightarrow u_1^2 + u_2^2 + u_3^2 + u_4^2 = 1$$

Ex - 5.1:

(03) Compute $\langle u, v \rangle$ using the inner product in Ex-7:

(a) $u = \begin{bmatrix} 3 & -2 \\ 4 & 8 \end{bmatrix}$, $v = \begin{bmatrix} -1 & 3 \\ 1 & 1 \end{bmatrix}$

(b) $u = \begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$, $v = \begin{bmatrix} 4 & 6 \\ 0 & 8 \end{bmatrix}$

$$\begin{aligned} \langle u, v \rangle &= \text{trace}(U^T V) \\ &= u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 \\ &= (3)(-1) + (-2)(3) + (4)(1) + 8(1) \\ &= -3 - 6 + 4 + 8 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \langle u, v \rangle &= 4 + 12 + 0 + 40 \\ &= 56 \end{aligned}$$

(13) Let M_{22} have the inner product in Ex-7. In each part find $\|A\|$.

a) $A = \begin{bmatrix} -2 & 5 \\ 3 & 6 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$\begin{aligned} \|A\| &= \langle A, A \rangle^{1/2} = \sqrt{u_1^2 + u_2^2 + u_3^2 + u_4^2} \\ &= \sqrt{4 + 25 + 9 + 36} \\ &= \sqrt{74} \end{aligned}$$

$$\|A\| = 0$$

(15) Let M_{22}

Ex-7. F

(a) $A = \begin{bmatrix} 2 \\ 9 \end{bmatrix}$

$d(A, B) =$

$A - B = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$

(b) $A = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$A - B = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$

(22) Let $U = \begin{bmatrix} u \\ v \end{bmatrix}$

$\langle U, V \rangle = 1$

on

$\langle u, v \rangle =$

Axiom-1:

$\langle u, v \rangle = 1$

Axiom-2:

$\langle u+v, w \rangle$

Axiom-4

But $\langle u,$

Axiom-4

-1+0+9+8

2x2 matrices

on M_{22}

-1+0+9+8

16

-7

$\begin{bmatrix} 4 & 6 \\ 0 & 8 \end{bmatrix}$

12+0+40

In

]

Let M_{22} have the inner product in

Ex-7. Find $d(A, B)$

① $A = \begin{bmatrix} 2 & 6 \\ 9 & 4 \end{bmatrix}, B = \begin{bmatrix} -4 & 7 \\ 1 & 6 \end{bmatrix}$

$d(A, B) = \langle A-B, A-B \rangle^{1/2} = \sqrt{(6)^2 + (-1)^2 + (8)^2 + (-2)^2}$

$A-B = \begin{bmatrix} 6 & -1 \\ 8 & -2 \end{bmatrix} = \sqrt{36+1+64+4} = \sqrt{105}$

② $A = \begin{bmatrix} -2 & 4 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -5 & 1 \\ 6 & 2 \end{bmatrix}$

$A-B = \begin{bmatrix} 3 & 3 \\ -5 & -2 \end{bmatrix} \quad d(A, B) = \langle A-B, A-B \rangle^{1/2} = \sqrt{9+9+25+4} = \sqrt{47}$

③ Let $U = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix}$ and $V = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$. Show that

$\langle U, V \rangle = u_1v_1 + u_2v_3 + u_3v_2 + u_4v_4$ is not an inner product on M_{22} .

Sol.

$\langle u, v \rangle = u_1v_1 + u_2v_3 + u_3v_2 + u_4v_4$ is not an inner product.

Axiom-1:

$\begin{aligned} \langle u, v \rangle &= u_1v_1 + u_2v_3 + u_3v_2 + u_4v_4 \\ &= v_1u_1 + v_3u_2 + v_2u_3 + v_4u_4 \\ &= \langle v, u \rangle \end{aligned}$

Axiom-3:

$\begin{aligned} \langle ku, v \rangle &= (ku)_1v_1 + \dots + (ku)_4v_4 \\ &= k[u_1v_1 + u_2v_3 + u_3v_2 + u_4v_4] \\ &= k\langle u, v \rangle \end{aligned}$

Axiom-2:

$\begin{aligned} \langle u+v, w \rangle &= (u_1+v_1)w_1 + (u_2+v_2)w_3 + (u_3+v_3)w_2 + (u_4+v_4)w_4 \\ &= (u_1w_1 + u_2w_3 + u_3w_2 + u_4w_4) + (v_1w_1 + v_2w_3 + v_3w_2 + v_4w_4) \\ &= \langle u, w \rangle + \langle v, w \rangle \end{aligned}$

Axiom-4:

~~$\langle u, u \rangle = u_1^2 + u_2^2 + u_3^2 + u_4^2$~~

But $\langle u, u \rangle = u_1^2 + u_2^2 + u_3^2 + u_4^2 \geq 0$ & $\langle u, u \rangle = 0$ if $u = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Note:

If $\langle u, v \rangle$ is the Euclidean inner product on $\mathbb{R}^n$, and if A is an $n \times n$ matrix, then

$$\langle u, Av \rangle = \langle A^T u, v \rangle.$$

25) Verify the above result for the Euclidean inner

product on $\mathbb{R}^3$ and $u = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & 0 \\ 5 & -1 & 2 \end{bmatrix}$

Solⁿ $\langle u, v \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3 = u \cdot v = v^T u$

L.H.S $\langle u, Av \rangle = (Av)^T u$

$$Av = \begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & 0 \\ 5 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 3+0-2 \\ 9+0+0 \\ 15+0-4 \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 11 \end{bmatrix}$$

$$\therefore \langle u, Av \rangle = (Av)^T u = \begin{bmatrix} 1 & 9 & 11 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = -1 + 18 + 44 = 61$$

R.H.S

$$\langle A^T u, v \rangle = (v)^T A^T u$$

$$A^T u = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & -1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -1+6+20 \\ 2+8-4 \\ -1+0+8 \end{bmatrix} = \begin{bmatrix} 25 \\ 6 \\ 7 \end{bmatrix}$$

$$\langle A^T u, v \rangle = (v)^T A^T u = \begin{bmatrix} 3 & 0 & -2 \end{bmatrix} \begin{bmatrix} 25 \\ 6 \\ 7 \end{bmatrix} = 75 + 0 - 14 = 61$$

$$L.H.S = R.H.S$$

Hence verified.

10. In each product $w = \langle u, v \rangle$

(a) the

$$\langle u, v \rangle$$

$$\|w\| =$$

(b) the vec where u

$$\|w\| =$$

(c) The in

$$A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\langle u, v \rangle =$$

$$\|w\| = \langle u, u \rangle$$

$$\|w\| = \langle u, u \rangle$$

on $\mathbb{R}^2$

In each part use the given inner product on $\mathbb{R}^2$ to find $\|w\|$, where $w = (1, 3)$.

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(a) the Euclidean inner product.

$$\langle u, v \rangle = u \cdot v = u_1 v_1 + u_2 v_2$$

$$\|w\| = \langle w, w \rangle^{1/2} = \sqrt{w_1^2 + w_2^2} = \sqrt{1 + 9} = \sqrt{10}$$

(b) the weighted Euclidean inner product $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$, where $u = (u_1, u_2)$ and $v = (v_1, v_2)$.

$$\|w\| = \langle w, w \rangle^{1/2} = \sqrt{3w_1^2 + 2w_2^2} = \sqrt{3 + 2(9)} = \sqrt{3 + 18} = \sqrt{21}$$

(c) The inner product generated by the matrix.

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$\langle u, v \rangle = v^T A^T A u = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -2 & 9 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} u_1 - 2u_2 \\ -2u_1 + 9u_2 \end{bmatrix} = (u_1 - 2u_2)v_1 + (-2u_1 + 9u_2)v_2$$

$$\langle u, u \rangle = u_1 u_1 - 2u_2 u_1 - 2u_1 u_2 + 9u_2 u_2$$

$$\|w\| = \langle w, w \rangle^{1/2} = \sqrt{w_1^2 - 2w_2 w_1 - 2w_1 w_2 + 9w_2^2}$$

$$= \sqrt{1^2 - 2(3)(-1) - 2(-1)(3) + 9(9)}$$

$$= \sqrt{1 + 6 + 6 + 81} = 4$$

$$= \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 13 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 2u_1 - u_2 \\ -u_1 + 13u_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= 2u_1 u_1 - u_1 u_2 - u_2 u_1 + 13u_2 u_2$$

$$\|w\| = \langle w, w \rangle^{1/2} = \sqrt{2w_1^2 - 2u_1 u_2 + 13u_2^2} = \sqrt{2 + 6 + 117}$$

$$= \sqrt{125}$$

inner

$$\begin{bmatrix} 1 & -2 & 1 \\ 3 & 4 & 0 \\ 5 & -1 & 2 \end{bmatrix}$$

8+44

0-14

11. Use the inner product in Exercise-10 to find $d(u, v)$ for $u = (-1, 2)$ and $v = (2, 5)$.

(a) $\langle u, v \rangle = u_1 v_1 + u_2 v_2$

$$\begin{aligned} d(u, v) &= \langle u - v, u - v \rangle^{1/2} = \langle (-3, -3), (-3, -3) \rangle^{1/2} \\ &= \sqrt{(-3)(-3) + (-3)(-3)} \\ &= \sqrt{18} = 3\sqrt{2} \end{aligned}$$

(b) $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$

$$\begin{aligned} d(u, v) &= \langle u - v, u - v \rangle^{1/2} = \sqrt{3(u_1 - v_1)(u_1 - v_1) + 2(u_2 - v_2)(u_2 - v_2)} \\ &= \sqrt{3(-3)(-3) + 2(-3)(-3)} \\ &= \sqrt{27 + 18} \\ &= \sqrt{45} = 3\sqrt{5} \end{aligned}$$

(c) $\langle u, v \rangle = 2u_1 v_1 - u_1 v_2 - u_2 v_1 + 13u_2 v_2$

$$\begin{aligned} d(u, v) &= \sqrt{2(u_1 - v_1)^2 - (u_1 - v_1)(u_2 - v_2) - (u_2 - v_1)(u_1 - v_2) + 13(u_2 - v_2)^2} \\ &= \sqrt{2(-3 - (-3))^2 - (-3 - 5)(-3 - 2) - (-3 - 2)(-3 - 5) + 13(-3 - 5)^2} \\ &= \sqrt{18 - 9 - 9 + 117} \\ &= \sqrt{117} = 3\sqrt{13} \end{aligned}$$

Example-8: An inner product on P_2 :

If $p = a_0 + a_1 x + a_2 x^2$ & $q = b_0 + b_1 x + b_2 x^2$ in P_2 ,

then $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$ is I.P on P_2

(Verify).

Note:

$$\|p\| = \langle p, p \rangle^{1/2} = \sqrt{a_0^2 + a_1^2 + a_2^2}$$

$$\text{if } \|p\| = 1 \Rightarrow a_0^2 + a_1^2 + a_2^2 = 1$$

Ex-6.1

(4) Compute in Examp

(a) $p = -2 + x$
 $\langle p, q \rangle =$

(b) $p = -5 + 2x$
 $\langle p, q \rangle = a_1$

(12) Let P_2 find $\|p\|$

(a) $p = -2 + 3x$

$$\begin{aligned} \|p\| &= \sqrt{(-2)^2 + 3^2} \\ &= \sqrt{4 + 9} \\ &= \sqrt{13} \end{aligned}$$

(14) Let P_2
 $d(p, q)$
 $p = 3 - x + 2x^2$

$$\langle p, q \rangle =$$

$$d(p, q) =$$

and $d(u, u)$

Ex-6.1

- ① Compute $\langle p, q \rangle$ using the inner product in Example 8.

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② $p = -2 + x + 3x^2$, $q = 4 - 7x^2$

$$\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2 = (-2)(4) + (1)(0) + (3)(-7) \\ = -8 - 21 = -29$$

③ $p = -5 + 2x + x^2$, $q = 3 + 2x - 4x^2$

$$\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2 = (-5)(3) + (2)(2) + (1)(-4) \\ = -15 + 4 - 4 = -15$$

- ④ Let P_2 have the inner product in Example-8. In each part find $\|p\|$.

① $p = -2 + 3x + 2x^2$

② $p = 4 - 3x^2$

$$\|p\| = \sqrt{(-2)^2 + (3)^2 + (2)^2} \\ = \sqrt{4 + 9 + 4} \\ = \sqrt{17}$$

$$\|p\| = \sqrt{16 + 9} = \sqrt{25} = 5$$

- ⑤ Let P_2 have the inner product in Example 8. Find $d(p, q)$.

$p = 3 - x + x^2$, $q = 2 + 5x^2$

$\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$ $p - q = 1 - x - 4x^2$

$$d(p, q) = \langle p - q, p - q \rangle^{1/2} = \sqrt{(1)^2 + (-1)^2 + (-4)^2} = \sqrt{2 + 16} \\ = \sqrt{18} \\ = 3\sqrt{2}$$

 $n P_2$ $n P_2$

Example-9: Inner product on $C[a, b]$:

Let $f = f(x)$ and $g = g(x)$ be two continuous functions in $C[a, b]$ and define, $\langle f, g \rangle = \int_a^b f(x)g(x)dx$ (1)

We P.T (1) is an I.P on $C[a, b]$.

Soln: $\langle f, g \rangle = \int_a^b f(x)g(x)dx = \int_a^b g(x)f(x)dx = \langle g, f \rangle$.

Axiom-1 satisfied.

Axiom-2:

$$\begin{aligned}\langle f+g, s \rangle &= \int_a^b [f(x)+g(x)] \cdot s(x) dx \\ &= \int_a^b f(x)s(x)dx + \int_a^b g(x)s(x)dx \\ &= \langle f, s \rangle + \langle g, s \rangle.\end{aligned}$$

~~Satisfy~~ holds true

Axiom-3:

$$\langle kf, g \rangle = \int_a^b (kf(x))g(x)dx = k \int_a^b f(x)g(x)dx = k \langle f, g \rangle.$$

holds true

Axiom-4:

If $f = f(x)$ is any function in $C[a, b]$, then

$$f^2(x) \geq 0 \quad \forall x \text{ in } [a, b]$$

$$\therefore \langle f, f \rangle = \int_a^b f^2(x)dx \geq 0.$$

Further, because $f^2(x) \geq 0$ and $f = f(x)$ is continuous on $[a, b]$, it follows that $\int_a^b f^2(x)dx = 0$ iff $f(x) = 0$ $\forall x$ in $[a, b]$.

Note:

$$\|f\| = \langle f, f \rangle^{1/2} = \sqrt{\int_a^b f^2(x)dx}$$

$$\Rightarrow \int_a^b f^2(x)dx = 1 \quad \text{iff} \quad \|f\| = 1. \quad \left(\text{Note: } \|f\|^2 = \langle f, f \rangle \right)$$

Theorem:

If u, v inner prod

a) $\langle 0, v \rangle$

c) $\langle u, ku \rangle$

e) $\langle u, u \rangle$

Ex-11: A88

Exercise: 1

17) Let the $\langle p, q \rangle =$

a) Find $\|p\|$

If $p=1$

If $p=x$, $\langle$

If $p=x^2$, $\langle$

b) $d(p, q)$ i

$d(p, q) =$

Ex. 11: $\langle u-2v, 3u+4v \rangle = \langle u, 3u+4v \rangle - \langle 2v, 3u+4v \rangle$
 $= \langle u, 3u \rangle + \langle u, 4v \rangle - \langle 2v, 3u \rangle - \langle 2v, 4v \rangle$
 $= 3\langle u, u \rangle + 4\langle u, v \rangle - 6\langle v, u \rangle - 8\langle v, v \rangle$
 $= 3\|u\|^2 + 4\langle u, v \rangle - 6\langle v, u \rangle - 8\|v\|^2$
 $= 3\|u\|^2 - 2\langle u, v \rangle - 8\|v\|^2$

(10)

Theorem - 6.1.1: Properties of I.P.s:

functions in

(1)

If u, v and w are vectors in a real inner product space, and ' k ' is any scalar, then:

- a) $\langle 0, v \rangle = \langle v, 0 \rangle = 0$ b) $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$
c) $\langle u, kv \rangle = k\langle u, v \rangle$ d) $\langle u-v, w \rangle = \langle u, w \rangle - \langle v, w \rangle$
e) $\langle u, v-w \rangle = \langle u, v \rangle - \langle u, w \rangle$

Ex-11: Assign

Exe - problem (16) Assign $\rightarrow$ in (A) page.
First do problem (28), (27).

Exercise: 6.1

17) Let the vector space P_2 have the inner product
 $\langle p, q \rangle = \int_{-1}^1 p(x)q(x) dx$

2) Find $\|p\|$ for $p=1$, $p=x$ and $p=x^2$.

If $p=1$, $\langle p, p \rangle = \sqrt{\int_{-1}^1 p^2 dx}$

$= \sqrt{\int_{-1}^1 1 dx} = \sqrt{(x)_{-1}^1} = \sqrt{2}$

If $p=x$, $\langle p, p \rangle = \sqrt{\int_{-1}^1 x^2 dx} = \sqrt{\left(\frac{x^3}{3}\right)_{-1}^1} = \sqrt{\frac{1}{3} + \frac{1}{3}} = \sqrt{\frac{2}{3}} = \frac{1}{\sqrt{3}}\sqrt{6}$

If $p=x^2$, $\langle p, p \rangle = \sqrt{\int_{-1}^1 x^4 dx} = \sqrt{\left(\frac{x^5}{5}\right)_{-1}^1} = \sqrt{\frac{1}{5} + \frac{1}{5}} = \sqrt{\frac{2}{5}} = \frac{1}{\sqrt{5}}\sqrt{10}$

8) $d(p, q)$ if $p=1$ and $q=x$.

$d(p, q) = \sqrt{\langle p-q, p-q \rangle} = \sqrt{\int_{-1}^1 (1-x)^2 dx}$

$= \sqrt{\int_{-1}^1 (1+x^2-2x) dx} = \sqrt{\left[x + \frac{x^3}{3} - \frac{2x^2}{2}\right]_{-1}^1}$

$= \sqrt{\left(1 + \frac{1}{3} - 1\right) - \left(-1 - \frac{1}{3} - 1\right)}$

$= \sqrt{2 + \frac{2}{3}} = \sqrt{\frac{8}{3}} = \frac{2}{\sqrt{3}}\sqrt{6}$

$k\langle f, g \rangle$

then

continuous

$f(x)=0$

27) Use the inner product $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$ to compute $\langle p, q \rangle$ for the vectors $p=p(x)$ and $q=q(x)$ in P_3 .

a) $p=1-x+x^2+5x^3$ $q=x-3x^2$

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$$

$$= \int_{-1}^1 (1-x+x^2+5x^3)(x-3x^2)dx$$

$$= \int_{-1}^1 (x-x^2+x^3+5x^4-3x^2+3x^3-3x^4-15x^5)dx$$

$$= \int_{-1}^1 (x-4x^2+4x^3+2x^4-15x^5)dx$$

$$= \left[\frac{x^2}{2} - \frac{4x^3}{3} + \frac{4x^4}{4} + \frac{2x^5}{5} - \frac{15x^6}{6} \right]_{-1}^1$$

$$= -\frac{4}{3}[1+1] + \frac{2}{5}[1+1]$$

$$= -\frac{8}{3} + \frac{4}{5} = \frac{-40+12}{15} = \underline{\underline{-\frac{28}{15}}}$$

b) $p=x-5x^3$, $q=2+8x^2$

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx = \int_{-1}^1 (x-5x^3)(2+8x^2)dx$$

$$= \int_{-1}^1 (2x-10x^3+8x^3-40x^5)dx$$

$$= \int_{-1}^1 (2x-2x^3-40x^5)dx$$

$$= \left[x^2 - \frac{2}{4}x^4 - \frac{40}{6}x^6 \right]_{-1}^1 = \underline{\underline{0}}$$

28) In each part use the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$ to compute $\langle f, g \rangle$ for the vectors $f=f(x)$ and $g=g(x)$ in $C[0, 1]$.

a) $f=\cos(2\pi x)$, $g=\sin(2\pi x)$ b) $f=x$, $g=e^x$ c) $f=\tan(\frac{\pi x}{4})$, $g=1$

$$\langle f, g \rangle = \int_0^1 \frac{1}{2} \sin(4\pi x) dx$$

$$= 0$$

$$= \int_0^1 x e^x dx$$

$$= (x e^x) - \int e^x dx$$

$$= [e - 0] - [e - 1] = \underline{\underline{1}}$$

$$\frac{2 \ln(2)}{\pi}$$

6.2:

Angle
product

Here a product of basic and a fundam and calc

Theorem - 6

If u and v are then

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

Ex - 1: Calc

Exercise - 6

a) In each part holds for

b) $u = (3, 2)$

$$\|u\| = \sqrt{9+4}$$

$$\|u\|$$

$$\langle u, v \rangle$$

$$v = (-3, 1, 0)$$

$$\|v\| = \sqrt{9+1}$$

$$\|v\| \|u\| = 11$$

$$\langle u, v \rangle = -6$$

$$|\langle u, v \rangle| = 6$$

$$\left[\ln(\sec \frac{\pi}{4}) \cdot \frac{4}{\pi} \right] = \frac{4}{\pi} \left[\ln(\sec \frac{\pi}{4}) - \ln \sec \frac{\pi}{4} \right] = \frac{4}{\pi} \ln \sqrt{2} = \frac{2}{\pi} \ln(2)$$

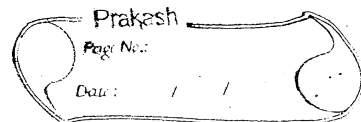
to compute

P_3 .

5) dx

6.2:

Angle and Orthogonality in Inner product spaces:



Here we define, an angle b/w two vectors in an inner product space, and use this concept to obtain some basic relations between vectors in an inner product, including a fundamental geometric relationship between the nullspace and column space of a matrix.

Theorem - 6.2.1: [Cauchy-Schwarz Inequality]

If u and v are vectors in a real inner product space, then

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

$$\because \cos \theta = \frac{u \cdot v}{\|u\| \|v\|} \quad \text{and } |\cos \theta| \leq 1 \Rightarrow \left| \frac{u \cdot v}{\|u\| \|v\|} \right| \leq 1 \Rightarrow |\langle u, v \rangle| \leq \|u\| \|v\|$$

Ex-1: (Assg)

Exercise - 6.2:

1) In each part verify that the Cauchy-Schwarz inequality holds for the given vectors. Using Euclidean I.P.

a) $u = (3, 2), v = (4, -1)$

$$\|u\| = \sqrt{9+4} = \sqrt{13} \quad \|v\| = \sqrt{16+1} = \sqrt{17}$$

$$\|u\| \|v\| = 14.866$$

$$\langle u, v \rangle = 3 \times 4 - 2 = 12 - 2 = 10 \quad \therefore |\langle u, v \rangle| \leq \|u\| \|v\|$$

b) $u = (-3, 1, 0), v = (2, -1, 3)$

$$\|u\| = \sqrt{9+1} = \sqrt{10}, \|v\| = \sqrt{4+1+9} = \sqrt{14}$$

$$\|u\| \|v\| = 11.83$$

$$\langle u, v \rangle = |-6 - 1 + 0| = 7$$

$$\Rightarrow |\langle u, v \rangle| \leq \|u\| \|v\|$$

c) $u = (-4, 2, 1), v = (8, -4, -2)$

$$\|u\| = \sqrt{16+4+1} = \sqrt{21}, \|v\| = \sqrt{64+16+4} = \sqrt{84} = 2\sqrt{21}$$

$$\|u\| \|v\| = 42$$

$$|\langle u, v \rangle| = |-32 - 8 - 2|$$

$$= 42$$

$$g = g(u) dx$$

$$g = g(u) \text{ in}$$

$$r(\frac{\pi x}{4}), g=1.$$

$$\ln(2)$$

$$\frac{1}{\pi}$$

27 Use the inner product $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$ to compute $\langle p, q \rangle$ for the vectors $p = p(x)$ and $q = q(x)$ in P_3

a) $p = 1 - x + x^2 + 5x^3$ $q = x - 3x^2$

$$\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$$

$$= \int_{-1}^1 (1 - x + x^2 + 5x^3)(x - 3x^2)dx$$

$$= \int_{-1}^1 (x - x^2 + x^3 - 3x^4 + 5x^4 - 15x^5)dx$$

$$= \int_{-1}^1 (x - x^2 + 2x^3 + 2x^4 - 15x^5)dx$$

$$= \int_{-1}^1 (x - x^2 + 2x^3 + 2x^4 - 15x^5)dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} + \frac{2x^4}{4} + \frac{2x^5}{5} - \frac{15x^6}{6} \right]_{-1}^1$$

$$= \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{2} + \frac{2}{5} - \frac{5}{2} \right) - \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{2} + \frac{2}{5} - \frac{5}{2} \right)$$

$$\|u\| \|v\| = 28.25 \quad |\langle u, v \rangle| = |-1 + 0 + 18 + 3| = 20$$

c) $p = -1 + 2x + x^2$; $q = 2 - 4x^2$ using IP $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$

$$\|p\| = \sqrt{0^2 + 2^2 + 1^2} = \sqrt{5}$$

$$= \sqrt{1 + 4 + 1} = \sqrt{6}$$

$$\|q\| = \sqrt{4 + 16} = \sqrt{20}$$

$$= \sqrt{20}$$

$$\|p\| \|q\| = 10.95 \quad |\langle p, q \rangle| = |-2 + 0 - 4| = 6$$

Can

Theorem-6.2.2: (Properties of length)

If u and v are vectors in an innerproduct space V and k is any scalar, then

a) $\|u\| \geq 0$ b) $\|u\| = 0$ iff $u = 0$ c) $\|ku\| = |k| \|u\|$

d) $\|u+v\| \leq \|u\| + \|v\|$ (Triangle inequality)

Theorem

If u , space V

a) $d(u$

d) $d(u,$

Angle

Let i then an $\cos$

Ex-2:

Let R^1 of the $u = ($

solve $\|u$

so that

Orthogo:

Two u orthogonal

Ex-3:

If M_{22}

then the

since $<$

Theorem 6.2.3: (Properties of distances)

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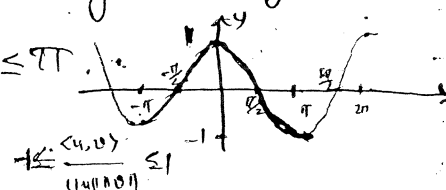
If u, v, w are vectors in an IP space V and k is any scalar, then

- (a) $d(u, v) \geq 0$ (b) $d(u, v) = 0$ iff $u = v$ (c) $d(u, v) = d(v, u)$
 (d) $d(u, v) \leq d(u, w) + d(w, v)$ (triangle inequality).

Angle b/w two vectors:

Let u and v be any two vectors in IP space V , then angle θ (unique) b/w u & v is given by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} \quad \text{and} \quad 0 \leq \theta \leq \pi$$



Ex-2:

Let $\mathbb{R}^4$ have the Euclidean innerproduct. Find the cosine of the angle θ between the vectors $u = (4, 3, 1, -2)$ & $v = (-2, 1, 2, 3)$

Soln: $\|u\| = \sqrt{20}$; $\|v\| = \sqrt{18}$ and $\langle u, v \rangle = -9$

So that $\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-9}{\sqrt{20} \sqrt{18}} = \frac{-3}{2\sqrt{15}} = -0.387$

Orthogonality:

Two vectors ' u ' and ' v ' in an IP space are called orthogonal if $\langle u, v \rangle = 0$.

Ex-3:

If M_{22} has the IP $\langle u, v \rangle = \text{trace}(UV) = u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4$

then the matrices $U = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ & $V = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ are orthogonal

since $\langle U, V \rangle = 0$.

5

$$+ 2u_2v_2$$

$$\text{trace}(U^T V)$$

$$u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4$$

$$= 20$$

$$= \int_0^b f(x) dx$$

$$= a_1b_1 + a_2b_2$$

space V

$$\|v\| = \|k\| \|u\|$$

Ex-4: Orthogonal vectors in P_2 :
Let P_2 have the IP $\langle p, q \rangle = \int_{-1}^1 p(x) q(x) dx$.

and let $p=x$ and $q=x^2$, then

$$\|p\| = \langle p, p \rangle^{1/2} = \left[\int_{-1}^1 x^2(x) dx \right]^{1/2} = \left[\int_{-1}^1 x^2 dx \right]^{1/2} = \left[\left[\frac{x^3}{3} \right]_{-1}^1 \right]^{1/2} \\ = \sqrt{\frac{2}{3}}.$$

$$\|q\| = \langle q, q \rangle^{1/2} = \left[\int_{-1}^1 x^4 dx \right]^{1/2} = \left[\left[\frac{x^5}{5} \right]_{-1}^1 \right]^{1/2} = \sqrt{\frac{2}{5}}.$$

$$\langle p, q \rangle = \int_{-1}^1 x^3 dx = \left[\frac{x^4}{4} \right]_{-1}^1 = \frac{1}{4} - \frac{1}{4} = 0.$$

$\therefore p$ and q are orthogonal relative to the given inner product.

~~Theorem~~: Theorem: Generalized Theorem of Pythagoras.

If u and v are orthogonal vectors in an inner product space, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

Eg: 5:

We knowed that $p=x$ and $q=x^2$ are orthogonal relative to the inner product

$$\langle p, q \rangle = \int_{-1}^1 p(x) q(x) dx \text{ on } P_2.$$

$$\|p+q\|^2 = \langle p+q, p+q \rangle = \int_{-1}^1 (x+x^2)(x+x^2) dx \\ = \int_{-1}^1 (x^2 + 2x^3 + x^4) dx \\ = \left[\frac{x^3}{3} + \frac{2x^4}{4} + \frac{x^5}{5} \right]_{-1}^1 \\ = \frac{2}{3} + 0 + \frac{2}{5} = \frac{16}{15}.$$

$$\|p\|^2 + \|q\|^2 = \frac{2}{3} + \frac{2}{5} = \frac{16}{15}.$$

Exercise

03) Let R_1^2
inner prod
cosine of

a) $u = (1, -3)$

$$\langle u, u \rangle =$$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

b) $u = (-1, 1)$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

c) $u = (4, 1, 8)$

$$\text{Ans} = \frac{-2}{9\sqrt{2}}$$

d) $u = (2, 1)$

$$v = (4, 0)$$

$$\text{Ans} =$$

e) let f
find

f) $p = -1$

$$\langle p, q \rangle =$$

$$\cos \theta = 0$$

g) $p = x - x$

$$\langle p, q \rangle =$$

$$\cos \theta =$$

Exercise: 6.2

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Let R^2, R^3 and R^4 have the Euclidean inner product. In each part find the cosine of the angle b/w u and v .

a) $u = (1, -3), v = (2, 4)$

$$\langle u, v \rangle = u_1 v_1 + u_2 v_2 = 2 - 12 = -10. \quad \|u\| = \sqrt{1+9} = \sqrt{10} \quad \|v\| = \sqrt{4+16} = \sqrt{20}$$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-10}{\sqrt{10} \cdot \sqrt{20}} = -0.707 \text{ or } -\frac{1}{\sqrt{2}}$$

b) $u = (-1, 0), v = (3, 8)$

$$\langle u, v \rangle = -3 \quad \|u\| = 1 \quad \|v\| = \sqrt{9+64} = \sqrt{73}$$

$$\cos \theta = \frac{-3}{\sqrt{73}}$$

c) $u = (-1, 5, 2)$

$v = (-2, 4, -9)$

$$\langle u, v \rangle = -2 + 20 - 18 = 0$$

$$\cos \theta = 0$$

d) $u = (4, 1, 8), v = (1, 0, -3)$

$$\text{Ans} = \frac{-20}{9\sqrt{10}}$$

e) $u = (1, 0, 1, 0); v = (-3, -3, -3, -3)$

$$\langle u, v \rangle = -3 + 0 - 3 + 0 = -6$$

$$\|u\| = \sqrt{2} \quad \|v\| = \sqrt{9+9+9+9} = \sqrt{36} = 6$$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-6}{\sqrt{2} \times 6} = -\frac{1}{\sqrt{2}}$$

f) $u = (2, 1, 7, -1)$

$v = (4, 0, 0, 0)$

$$\text{Ans} = 2/\sqrt{55}$$

4) Let P_2 have the IP $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$. Find the cosine of the angle b/w p and q .

a) $p = -1 + 5x + 2x^2, q = 2 + 4x - 9x^2$

$$\langle p, q \rangle = -2 + 20 - 18 = 0$$

$$\cos \theta = 0$$

b) $p = x - x^2, q = 7 + 3x + 3x^2$

$$\langle p, q \rangle = 0 + 3 - 3 = 0$$

$$\cos \theta = 0$$

06) Let M_{22} have $\text{I.P.} \langle u, v \rangle = \text{trace}(u^t v)$. Find the cosine of the angle b/w A & B .

② $A = \begin{bmatrix} 2 & 6 \\ 1 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

$$\langle A, B \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

$$= 6 + 12 + 1 - 0 = 19.$$

$$\|A\| = \sqrt{4 + 36 + 1 + 9} = \sqrt{50} \quad \|B\| = \sqrt{9 + 4 + 1 + 0} = \sqrt{14}.$$

$$\cos \theta = \frac{\langle A, B \rangle}{\|A\| \|B\|} = \frac{19}{\sqrt{50} \cdot \sqrt{14}} = 0.718 \text{ or } \frac{19}{10\sqrt{7}}.$$

③ $A = \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$

$$\langle A, B \rangle = -6 + 4 - 4 + 6 = 0.$$

$$\cos \theta = 0.$$

07) In each part determine whether the given vectors are orthogonal w.r.t the Euclidean I.P.

① $u = (-1, 3, 2)$, $v = (4, 2, -1)$

$$\langle u, v \rangle = -4 + 6 - 2 = 0$$

Yes

② $u = (-2, -2, -2)$

$v = (1, 1, 1)$

No

③ $u = (u_1, u_2, u_3)$, $v = (0, 0, 0)$ (Yes)

④ $u = (-4, 6, -10, 1)$
 $v = (2, 1, -2, 9)$

No

⑤ $u = (0, 3, -2, 1)$, $v = (5, 2, -1, 0)$ No

⑥ $u = (a, b)$, $v = (-b, a)$

Yes

08) Show that $p = 1 - x + 2x^2$ and $q = 2x + x^2$ are orthogonal with respect to the inner product $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$.

Soln:-

$$\langle p, q \rangle = 0 - 2 + 2 = 0.$$

Yes.

07) Let $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ following w.r.t

② $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

$$\langle A, B \rangle = \text{Yes}$$

08) Let R^3 are -u

② $u = (2, 1, 1)$

$$\langle u, u \rangle =$$

Orthogonal

Let W

u in V

orthogonal

all vects

called

denoted

Thm-6.6

If W

space V .

② $W^\perp$ is

③ The

④ The

($W^\perp$)

Find

Let $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, which of the

following matrices are orthogonal

w.r.t to IP $\langle u, v \rangle = \text{trace}(U^T V) = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$.

(A) $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

$\langle A, B \rangle = -6 + 0 + 0 + 6 = 0$

Yes

(B) $\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$

Yes

(C) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Yes

(D) $\begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$

No.

$+1+0 = \sqrt{14}$

$\frac{19}{10\sqrt{7}}$

Q8) Let $\mathbb{R}^3$ have the Euclidean IP. For which values of k are u and v orthogonal?

(A) $u = (2, 1, 3), v = (1, 7, k)$

(B) $u = (k, k, 1), v = (k, 5, 6)$

$\langle u, v \rangle = 0 \Rightarrow 2 + 7 + 3k = 0$

$\langle u, v \rangle = 0 \Rightarrow k^2 + 5k + 6 = 0$

$3k = -9$

$k = -3$

$k = -2, -3$

vectors are

Orthogonal Complements:

Let W be a subspace of an IP space V . A vector u in V is said to be orthogonal to W if it is orthogonal to every vector in W , and the set of all vectors in V that are orthogonal to W is called the orthogonal complement of W , and it is denoted by $W^\perp$ (W perp).

Thm-6.2.5: Properties of orthogonal complements.

If W is a subspace of a finite dimensional inner product space V , then:

(A) $W^\perp$ is a subspace of V .

(B) The only vector common to W and $W^\perp$ is 0 .

(C) The orthogonal complement of $W^\perp$ is W ; that is

$(W^\perp)^\perp = W$.

$v = (-b, a)$

Yes

orthogonal

$a_0 b_0 + a_1 b_1$

$+ a_2 b_2$

9)

No

A Geometric Link between Nullspace and Row Space

Theorem - 6.2.6:

If A is an $m \times n$ matrix, then:

- (a) The nullspace of A and the row space of A are orthogonal complements in $\mathbb{R}^n$ with respect to the Euclidean inner product.
- (b) The nullspace of A^T and the column space of A are orthogonal complements in $\mathbb{R}^m$ with respect to the Euclidean inner product.

Note:

The orthogonal complement of the row space of A is the nullspace of A :

is we must show that, if $AV=0$, then v is orthogonal to every vector in the row space and conversely, if a vector v is orthogonal to every vector in the row space, then $AV=0$.

Ex-6: Basis for an Orthogonal Complement:

Let W be the subspace of $\mathbb{R}^5$ spanned by the vectors $w_1 = (2, 2, -1, 0, 1)$; $w_2 = (-1, -1, 2, -3, 1)$

$$w_3 = (1, 1, -2, 0, -1), \quad w_4 = (0, 0, 1, 1, 1).$$

Find a basis for the orthogonal complement of W .

Soln:- The space W spanned by the vectors w_1, w_2, w_3 and w_4 is the same as the Row space of the matrix

$$A = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

of A
of W
 $\therefore A =$

$$\begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 =$$

$$(f) \|u - 2v + 4w\|^2 = \langle u - 2v + 4w, u - 2v + 4w \rangle = \langle u, u - 2v + 4w \rangle - \langle 2v, u - 2v + 4w \rangle + \langle 4w, u - 2v + 4w \rangle$$

$$= \langle u, u \rangle - \langle u, 2v \rangle + \langle u, 4w \rangle - \langle 2v, u \rangle + \langle 2v, 2v \rangle - \langle 2v, 4w \rangle + \langle 4w, u \rangle - \langle 4w, 2v \rangle + \langle 4w, 4w \rangle$$

$$= \|u\|^2 - 4\langle u, v \rangle + 4\|w\|^2 + 16\|w\|^2 + 8\langle u, w \rangle - 16\langle v, w \rangle$$

$$= 1 + 4 \times 4 + 16 \times 49 - 4 \times 2 + 8 \times 5 - 16 \times 3 = 1 + 16 + 784 - 8 + 40 + 48 = 881$$

Row Space

from theorem - 6.2.6 @, the nullspace of A is the orthogonal complement of W .

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-1 + 4.

Space of A is orthogonal to

$$A = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_1 + 2R_2$$

$$R_3 \rightarrow R_1 - 2R_3$$

$$-1 + 4.$$

Space of A is orthogonal to the

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

of A is

$$\rightarrow \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is orthogonal

exactly,

to in the

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

$$x_1 = -x_2 - x_3$$

$$x_4 = 0, x_5 = -x_3$$

$$x_1 = -x_2 - x_3, x_4 = 0, x_5 = -x_3$$

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

by the

-3, 1)

$\therefore$ Basis for orthogonal complement of W is $\{v_1, v_2\}$.

Ex (16): Suppose that u, v , and w are vectors such that

$$\langle u, v \rangle = 2, \langle v, w \rangle = -3, \langle u, w \rangle = 5, \|u\| = 1, \|v\| = 2, \|w\| = 7$$

Evaluate the given expression

$$(a) \langle u+v, v+w \rangle \quad (b) \langle 2v-w, 3u+2w \rangle \quad (c) \langle u-v-2w, 4u+v \rangle$$

$$(d) \|u+v\| \quad (e) \|2w-v\| \quad (f) \|u-2v+4w\|$$

$$\text{Sol (a)} \langle u, v+w \rangle + \langle v, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle + \langle v, v \rangle + \langle v, w \rangle = 2 + 5 + 4 - 3 = 8$$

$$(b) \langle 2v, 3u+2w \rangle + \langle w, 3u+2w \rangle = \langle 2v, 3u \rangle + \langle 2v, 2w \rangle + \langle w, 3u \rangle + \langle w, 2w \rangle$$

$$= 6\langle v, u \rangle + 4\langle v, w \rangle - 3\langle w, u \rangle - 2\langle w, w \rangle = 12 - 12 - 15 - 2 \times 49 = -113$$

$$(c) \langle u, 4u+v \rangle - \langle v, 4u+v \rangle - \langle 2w, 4u+v \rangle = \langle u, 4u \rangle + \langle u, v \rangle - \langle v, 4u \rangle - \langle v, v \rangle - \langle 2w, 4u \rangle - \langle 2w, v \rangle$$

$$= 4\langle u, u \rangle + \langle u, v \rangle - 4\langle v, u \rangle - \langle v, v \rangle - 8\langle w, u \rangle - 2\langle w, v \rangle$$

$$= 4 \times 1 + 2 - 4 \times 2 - 4 - 8 \times 5 - 2 \times (-3) = 4 - 4 - 8 - 10 + 6 = -12$$

Exercise - 6.2

02 Let $\mathbb{R}^4$ have the Euclidean IP and $u = (-1, 1, 0, 2)$.
Determine whether the vector 'u' is orthogonal to
the subspace spanned by the vectors $w_1 = (0, 0, 0, 0)$
 $w_2 = (1, -1, 3, 0)$ $w_3 = (4, 0, 9, 2)$.

Sol The subspace W spanned by the vectors w_1, w_2 & w_3
is the same as the row space of the matrix,

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & -1 & 3 & 0 \\ 4 & 0 & 9 & 2 \end{bmatrix}$$

To check whether 'u' is orthogonal to the subspace
 W , we need to show that $Au = 0$.

(i.e. 'u' is the nullspace of A)

From Thm-6.2.6 i.e., we will show that vector 'u'
is orthogonal to every vector in the row space.

$$\therefore \langle u, w_1 \rangle = 0, \langle u, w_2 \rangle = -2; \langle u, w_3 \rangle = 0.$$

Since u is not orthogonal to the row vector w_2 ,
we conclude 'u' is not orthogonal to the subspace W .

12 Let W be the line in $\mathbb{R}^2$ with the equation
 $y = 2x$. Find an eqn for $W^\perp$.

$$y = 2x$$

Sol

$$2x - y = 0.$$

$$\begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Sol is is $x = t$ $y = 2t$
Substitute $\begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} t \\ 2t \end{bmatrix} = 0$
 $\Rightarrow 2t - 2t = 0$

$l = (-1, 1, 0, 2)$
 genal to
 $z = (0, 0, 0, 0)$

u_1, u_2 & u_3
 matrix,

the subspace

vectors in
 no space.

$z > 0$

cos u_2 ,
 subspace W ,

uation.

$22x$

$W = \{ (x, y) \text{ which satisfy the}$
 eqⁿ $[2 \ -1] \begin{bmatrix} x \\ y \end{bmatrix} = 0$.

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$\therefore W^\perp$ is the row space of A i.e. $\vec{v} = [2 \ -1]$
 Parametric eqⁿ of $W^\perp$, $t\vec{v}$.
 ~~$2t$ $y=0$~~ $x=2t \ y=t$

or
 $m_1 m_2 = -1$

$2 m_2 = -1$
 $m_2 = -\frac{1}{2}$

Eqⁿ of $W^\perp \rightarrow$

$$\begin{aligned} x &= -2y \\ y &= -\frac{1}{2}x \end{aligned}$$

$\Rightarrow y = -\frac{1}{2}x$

13) Let W be the plane in R^3 with eqⁿ $x - 2y - 3z = 0$.
 Find the parametric eq^{ns} for $W^\perp$.

Ans $[1 \ -2 \ -3] \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

W is the plane that contains pts (x, y, z) which satisfies
 the above eqⁿ is W is the nullspace of the above eqⁿ.

$\therefore$ The orthogonal complement $W^\perp$ of W is the row
 space of the matrix, is $u = [1 \ -2 \ -3]$

The parametric equations for $W^\perp$ is $= t\vec{v}$.
 (any line $\parallel$ to normal is $t\vec{v}$) $= \begin{bmatrix} t & -2t & -3t \\ x & y & z \end{bmatrix}$
 $\Rightarrow x = t \quad y = -2t \quad z = -3t$

The eqⁿ of $W^\perp$ is $y = -2x \quad z = -3x$

$y + 2x = 0$
 $3x + z = 0$

$x - y + z = 0$

b) Let W be the line in $\mathbb{R}^3$ with the parametric eqns $x=2t, y=-5t, z=4t$ ($-\infty < t < \infty$).

Find an eqn for $W^\perp$. $\downarrow$ eqn of a line $\parallel$ to $(2, -5, 4)$.

Solve $[2 \ -5 \ 4]t = 0.$

W is the row space of the matrix given.
The orthogonal complement is set of all solutions which satisfy this eqn. is $[2 \ -5 \ 4] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

$$\Rightarrow 2x - 5y + 4z = 0. \Rightarrow \text{plane}$$

~~is eqn of a line.~~

© Let W be the intersection of two planes $x+y+z=0, x-y+z=0$ in $\mathbb{R}^3$. Find the eqns for $W^\perp$.

Solve Solving the two eqns we get $y=0$ (subtract).
 $x+z=0.$

$\therefore W$ is a line in xz -plane with the eqn $x+z=0.$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

$\therefore$ The $W^\perp$ is the row space of the above eqn.

$$\vec{w} = [1 \ 0 \ 1]$$

The parametric eqn is $t\vec{w} = [t \ 0 \ t]$

$$\text{is } x=t \quad y=0 \quad z=t$$

$$\Rightarrow x=z$$

$$\Rightarrow x-z=0. \text{ is a line in } xz \text{ plane}$$

This is the eqn of the $W^\perp$.

14. Let

Find b
Verify th

Solve

$$\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$\therefore$ The

$\therefore$

$\therefore$ let

$$X =$$

$\therefore$ The

$$\langle \gamma, X, \gamma \rangle$$

$$\langle \gamma, X, \gamma \rangle$$

parametric
 $t \in (-\infty, \infty)$

11 to
 $(-5, 4)$

given
 solutions

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

0. $\Rightarrow$ plane

planes
 the

(Subtract)

2.

$$x + z = 0$$

one eqn.

in xz plane

Let $A = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 5 & 0 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix}$

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 (Thm 6.2.6)

Find bases for row space and Null space of A .
 Verify that every vector in the row space is orthogonal to vectors in the N.S.
 $R_2 \rightarrow R_2 - 3R_1$
 $R_3 \rightarrow R_3 - R_1$

$$\begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 5 & 0 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & -1 & 3 & -2 \\ 0 & -1 & 3 & -2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & -1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ The basis for row space is $(1, 2, -1, 2)$,
 $(0, 1, -3, 2)$.

$$A \cdot X = 0 \Rightarrow \begin{aligned} x_1 + 2x_2 - x_3 + 2x_4 &= 0 \\ x_2 - 3x_3 + 2x_4 &= 0 \end{aligned}$$

$$\begin{aligned} x_2 &= 3x_3 - 2x_4 & x_1 &= x_3 - 2(3x_3 - 2x_4) \\ & & &= x_3 - 6x_3 + 4x_4 \\ & & &= -5x_3 + 4x_4 \end{aligned}$$

let $x_3 = t$ $x_4 = r$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -5t + 2r \\ 3t - 2r \\ t \\ r \end{bmatrix} = t \begin{bmatrix} -5 \\ 3 \\ 1 \\ 0 \end{bmatrix} + r \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ The basis for Nullspace is $X_1 = \begin{bmatrix} -5 \\ 3 \\ 1 \\ 0 \end{bmatrix}$, $X_2 = \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}$

$$\langle r_1, X_1 \rangle = 0, \quad \langle r_2, X_1 \rangle = 0, \quad \langle r_3, X_1 \rangle = 0$$

15. Let A be the matrix in (14).

(a) Find bases for the column space of A and null space of A^T .

(b) Verify that every vector in the column space of A is orthogonal to every vector in the null space of A^T (6.2.6 (b)).

Solve

$$A = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 5 & 0 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The leading 1's in Row reduced echelon form matrix R are $C_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ & $C_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

The corresponding column vectors in A are

$$C_1 = (1, 3, 1)^T \quad C_2 = (2, 5, 1)^T$$

$\therefore$ The basis for column space is $\{C_1, C_2\}$
 $= \{(1, 3, 1)^T, (2, 5, 1)^T\}$

$$A^T = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 1 \\ -1 & 0 & 2 \\ 2 & 4 & 0 \end{bmatrix} \approx \begin{bmatrix} 1 & 3 & 1 \\ 0 & -1 & -1 \\ 0 & 3 & 3 \\ 0 & -2 & -2 \end{bmatrix} \quad 4-6$$

$$= \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 3x_2 + x_3 = 0$$

$$x_2 + x_3 = 0$$

$$x_2 = -x_3$$

~~4-2~~

$$x_1 - 3x_3 + x_3 = 0$$

$$x_1 = 2x_3$$

$$\therefore X = t \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

(16) Find a subspace

(a) $U_1 = \{$

$$A \times 20$$

(b) $U_1 = (2, 1, -1)^T$

(c) $U_1 = (1, 4, -1)^T$

(d) $U_1 = (1, 4, -1)^T$

$$(-1, -1)$$

4 and

space of
null space

$$\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$= 2 - 3 + 1 = 0$$

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} = 4 - 5 + 1 = 0$$

$$\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -4 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \end{bmatrix} = -2 + 0 + 2 = 0$$

$$\begin{bmatrix} -1 & 2 \\ -3 & 2 \\ 0 & 0 \end{bmatrix}$$

on form

}

we

16) Find a basis for the orthogonal complement of the subspace spanned by the vectors. $A = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

a) $v_1 = (1, -1, 3)$ $v_2 = (5, -4, -4)$ $v_3 = (7, -6, 2)$

$A \times 20 \Rightarrow$ Basis Sol $(16, 19, 1)$

b) $v_1 = (2, 0, -1)$ $v_2 = (4, 0, -2)$ $(0, 1, 0)$ $(1/2, 0, 1)$

c) $v_1 = (1, 4, 5, 2)$ $v_2 = (2, 1, 3, 0)$ $v_3 = (-1, 3, 2, 2)$
 $(-1, -1, 1, 0)$ $(2/7, -4/7, 0, 1)$

d) $v_1 = (1, 4, 5, 6, 9)$ $(3, -2, 1, 4, -1)$ $v_2 = (-1, 0, -1, -2, -1)$
 $v_4 = (2, 3, 5, 7, 8)$
 $(-1, -1, 1, 0, 0)$ $(-2, -1, 0, 1, 0)$ $(-1, -2, 0, 0, 1)$

e) $\begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$ If $x_3 = 19$, then
 $x_1 = 16$; $x_2 = 19$
 $X = \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix}$

$-3x_3 + x_2 = 0$

$9x_1$

09) Let $\mathbb{R}^4$ have the Euclidean I.P. Find two vectors of room 1, that are orthogonal to the three vectors $u = (2; 1; 4, 0)$, $v = (-1, -1, 2, 2)$ $w = (3, 2, 5, 4)$.

$$\begin{bmatrix} 2 & 1 & -4 & 0 \\ -1 & -1 & 2 & 2 \\ 3 & 2 & 5 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = 0.$$

Find (u_1, u_2, u_3, u_4) s.t. $\|u\| = 1$.

$$\begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \begin{bmatrix} 2 & 1 & -4 & 0 \\ 0 & -1 & 0 & 4 \\ 0 & 1 & 22 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & -4 & 0 \\ 0 & -1 & 0 & 4 \\ 0 & 0 & 22 & 12 \end{bmatrix}$$

$$2u_1 + u_2 - 4u_3 = 0$$

$$-u_2 + 4u_4 = 0$$

$$22u_3 + 12u_4 = 0$$

$$u_2 = 4u_4$$

$$u_3 = -\frac{12}{22}u_4 = -\frac{6}{11}u_4$$

$$2u_1 = 4u_3 - 2u_2$$

$$2u_1 = 4 \times -\frac{6}{11}u_4 - 4u_4 = -\frac{24}{11}u_4 - \frac{44}{11}u_4 = -\frac{68}{11}u_4$$

$$2u_1 = -\frac{24u_4 - 44u_4}{11}$$

$$2u_1 = -\frac{68}{11}u_4$$

$$u_1 = -\frac{34}{11}u_4$$

$$u_2 = 4u_4; u_3 = -\frac{6}{11}u_4; u_1 = -\frac{34}{11}u_4$$

Let $u_4 = 11$.

$$\Rightarrow u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -34 \\ 44 \\ -6 \\ 11 \end{bmatrix}$$

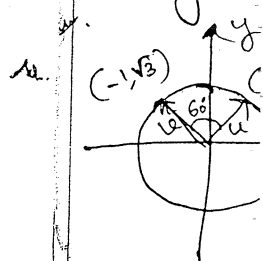
Given $\|u\| = 1$.

$$\sqrt{3249} = 57$$

we need two vectors of room 1.

$$\therefore u = \pm \frac{1}{57} (-34, 44, -6, 11)$$

30) With reg product, $v = (-1, \sqrt{3})$ bln them Euclidean orthogonal



$$\|u\|^2 = 1$$

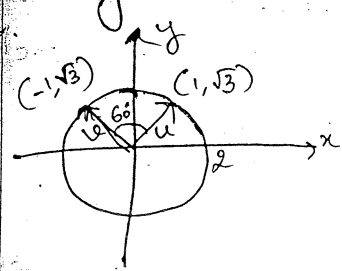
solving (

∴ The u

find two
normal to
 $= (-1, -1, 2)$

With respect to the Euclidean inner product, the vectors $u = (1, \sqrt{3})$ and $v = (-1, \sqrt{3})$ have norm 2, and the angle b/w them is 60° (see accompanying figure). Find a weighted Euclidean inner product w.r.t which u & v are orthogonal unit vectors.

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$\langle u, v \rangle = 0$ for them to be orthogonal & unit vectors $\Rightarrow \|u\|^2 = 1, \|v\|^2 = 1$.

$$\therefore \langle u, v \rangle = w_1 u_1 v_1 + w_2 u_2 v_2$$

$$\Rightarrow \langle u, v \rangle = 0 \Rightarrow -w_1 + 3w_2 = 0 \rightarrow (1)$$

$$\|u\|^2 = 1, \|v\|^2 = 1 \Rightarrow w_1 + 3w_2 = 1 \rightarrow (2)$$

Solving (1) & (2) we get $w_2 = \frac{1}{6}, w_1 = \frac{1}{2}$.

$\therefore$ The weighted Euclidean inner product is $\langle u, v \rangle = w_1 u_1 v_1 + w_2 u_2 v_2$
 $= \frac{1}{2} u_1 v_1 + \frac{1}{6} u_2 v_2$

$$\begin{bmatrix} 2 & 1 & -4 \\ 0 & -1 & 0 \\ 0 & 0 & 22 \end{bmatrix}$$

$$= 4u_3 - 2u_2$$

$$= \frac{4x - 6u_4}{11}$$

$$\frac{4u_4 - 44u_4}{11}$$

$$\frac{18}{11} u_4$$

$$\frac{34}{11} u_4$$

6.3:

Orthonormal Bases; Gram-Schmidt process;

QR-Decomposition:

Defn:

A set of vectors in an inner product space is called an orthogonal set if all pairs of distinct vectors in the set are orthogonal. An orthogonal set in which each vector has norm 1 is called orthonormal.

Example-1:

Let $u_1 = (0, 1, 0)$; $u_2 = (1, 0, 1)$; $u_3 = (1, 0, -1)$ and assume that $\mathbb{R}^3$ has the Euclidean inner product. It follows that the set of vectors $S = \{u_1, u_2, u_3\}$ is orthogonal since $\langle u_1, u_2 \rangle = \langle u_1, u_3 \rangle = \langle u_2, u_3 \rangle = 0$.

Note:

If v is a nonzero vector in an IP space, then $\frac{v}{\|v\|}$ is a vector of norm 1.

Constructing Orthonormal Set:

Exr: The Euclidean norm of the vectors in Example 1 are $\|u_1\| = 1$; $\|u_2\| = \sqrt{2}$, $\|u_3\| = \sqrt{2}$.

$$\therefore v_1 = \frac{u_1}{\|u_1\|} = (0, 1, 0), \quad v_2 = \frac{u_2}{\|u_2\|} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right),$$

$$v_3 = \frac{u_3}{\|u_3\|} = \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right) \text{ are of unit norm.}$$

$\Rightarrow \{v_1, v_2, v_3\}$ is a orthonormal set.

do exercise problems:

Defn:

In

Constructing

Called

Construct

basis.

Ex: 1

Exercise

01) which of with or

(a) $(0, 1)$,

(c) $(-\sqrt{2}, -1)$

02) Which of the Eu

03) Which with or

(a) $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

04) $\langle u_1, u_2 \rangle$

No.

(b) $\left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right)$

$\langle u_1, u_2 \rangle =$

$\langle u_2, u_3 \rangle =$

$\langle u_1, u_3 \rangle =$

Defn:

In an IP space, a basis consisting of orthonormal vectors is called an orthonormal basis for $\mathbb{R}^3$. And a basis consisting of orthogonal vectors is called an orthogonal basis.

Ex: i, j, k are orthonormal basis for $\mathbb{R}^3$.

Exercise - 6.3:

1) and assume it follows is orthogonal

space, then

Example 1

2)

sum

1) which of the following sets of vectors are orthogonal with respect to the Euclidean inner product on $\mathbb{R}^2$?

- (a) $(0,1), (2,0)$ ✓ (b) $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ ✓
 (c) $(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (d) $(0,0), (0,1)$ ✓

2) Which of the sets in Exercise 1 are orthonormal w.r.t the Euclidean inner product on $\mathbb{R}^2$? (b)

3) Which of the following sets of vectors are orthogonal with respect to the Euclidean inner product on $\mathbb{R}^3$?

- (a) $(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}), (-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$

Sol $\langle u_1, u_2 \rangle = (\frac{1}{\sqrt{2}} + 0 + \frac{1}{\sqrt{2}}) \neq 0$ $\langle u_2, u_3 \rangle = -\frac{1}{\sqrt{2}} + 0 + \frac{1}{\sqrt{2}} = 0$

No.

- (b) $(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}), (\frac{2}{3}, \frac{1}{3}, \frac{2}{3}), (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

$\langle u_1, u_2 \rangle = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$

$\langle u_2, u_3 \rangle = \frac{2}{9} + \frac{2}{9} - \frac{4}{9} = 0$

$\langle u_1, u_3 \rangle = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0$

$\therefore$ The given set of vectors are orthogonal w.r.t EIP.

c) $(1, 0, 0), (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (0, 0, 1)$.

$\langle u_1, u_2 \rangle = 0$; $\langle u_2, u_3 \rangle = \frac{1}{\sqrt{2}} \neq 0$; $\langle u_1, u_3 \rangle = 0$

No

d) $(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}) \cdot (\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0) = \text{Yes}$

04) Which of the sets in Exercise-3 are orthonormal with respect to the Euclidean inner product on $\mathbb{R}^3$?

(b) & (d)

05) Which of the following sets of polynomials are orthonormal w.r.t the inner product on P_2 discussed in Ex-8 of Section 6.1? $\langle u, v \rangle = \int_{-1}^1 uv dx$

a) $\frac{2}{3} - \frac{2x}{3} + \frac{x^2}{3}$; $\frac{2}{3} + \frac{x}{3} - \frac{2}{3}x^2$; $\frac{1}{3} + \frac{2x}{3} + \frac{2x^2}{3}$

Soln. $\langle p_1, p_2 \rangle = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$; $\langle p_2, p_3 \rangle = \frac{2}{9} + \frac{2}{9} - \frac{4}{9} = 0$

$\langle p_1, p_3 \rangle = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0$

$\therefore$ The given set of polynomials are orthogonal.

$\|p_1\| = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = \sqrt{1} = 1$; $\|p_2\| = 1$; $\|p_3\| = 1$

$\therefore S$ is orthonormal set.

b) 1 ; $\frac{x}{\sqrt{2}} + \frac{x^2}{\sqrt{2}}$; x^2

Soln $\langle p_1, p_2 \rangle = 0$; $\langle p_2, p_3 \rangle = 0 + 0 + \frac{1}{\sqrt{2}} \neq 0$; $\langle p_1, p_3 \rangle = 0$

$\therefore$ The given set of polynomials are not orthogonal and hence it is not an orthonormal set.

06) Which of the following sets of matrices are orthonormal w.r.t IP on M_{22} $\langle U, V \rangle = u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4$

a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 2/3 \\ 1/3 & -2/3 \end{bmatrix}$, $\begin{bmatrix} 0 & 2/3 \\ -2/3 & 1/3 \end{bmatrix}$, $\begin{bmatrix} 0 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}$ Yes

b) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

07) Verify the the Euclidean orthonormal

a) $(-1, 2)$; $(1, 1)$

$\langle u_1, u_2 \rangle =$

$\|u_1\| = \sqrt{1}$

$\therefore$ The

b) $(1, 0, -1)$

$\langle u_1, u_2 \rangle =$

$\|u_1\| = \sqrt{2}$

$\therefore$ The

$(\frac{1}{\sqrt{2}}, 0)$

c) $(\frac{1}{5}, \frac{1}{5})$

$\langle u_1, u_2 \rangle =$

$\|u_1\| = \sqrt{2}$

$\|u_3\| = \sqrt{2}$

The 0

$(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$

$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

⑤ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}$ (No)

07) Verify that the given set of vectors is orthogonal w.r.t the Euclidean inner product; then convert it to an orthonormal set by normalizing the vectors.

a) $(-1, 2); (6, 3)$

$$\langle u_1, u_2 \rangle = -6 + 6 = 0$$

$$\|u_1\| = \sqrt{1+4} = \sqrt{5} \quad \|u_2\| = \sqrt{36+9} = \sqrt{45} = 3\sqrt{5}$$

$\therefore$ The orthonormal set of vectors are $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right), \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

⑥ $(1, 0, -1); (2, 0, 2); (0, 5, 0)$

$$\langle u_1, u_2 \rangle = 2 - 2 = 0; \quad \langle u_2, u_3 \rangle = 0; \quad \langle u_1, u_3 \rangle = 0$$

$$\|u_1\| = \sqrt{2}; \quad \|u_2\| = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}; \quad \|u_3\| = \sqrt{25} = 5$$

$\therefore$ The orthonormal set of vectors are $\left\{\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), (0, 1, 0)\right\}$

⑦ $\left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right), \left(-\frac{1}{2}, \frac{1}{2}, 0\right), \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right)$

$$\langle u_1, u_2 \rangle = -\frac{1}{10} + \frac{1}{10} = 0 \quad \langle u_2, u_3 \rangle = -\frac{1}{6} + \frac{1}{6} = 0 \quad \langle u_1, u_3 \rangle = \frac{1}{15} + \frac{1}{15} - \frac{2}{15} = 0$$

$$\|u_1\| = \sqrt{\frac{1}{25} + \frac{1}{25} + \frac{1}{25}} = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}; \quad \|u_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 0} = \frac{\sqrt{2}}{2}$$

$$\|u_3\| = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{6}{9}} = \frac{\sqrt{6}}{3}$$

$\therefore$ The orthonormal set of vectors are

$$\left\{\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)\right\}$$

Orthonormal

$$u_3 + u_4 u_4$$

08) Verify that the set of vectors $\{(1,0), (0,1)\}$ is orthogonal w.r.t the inner product $\langle u,v \rangle = 4u_1v_1 + u_2v_2$ on $\mathbb{R}^2$; then convert it to an orthonormal set by normalizing the vectors.

Soln

$$\langle u_1, u_2 \rangle = 4 \times 0 + 0 = 0.$$

$\therefore \{(1,0), (0,1)\}$ is orthogonal w.r.t the IP given.

$$\|u_1\| = \sqrt{4u_1^2 + u_2^2} = \sqrt{4} = 2 \quad ; \quad \|u_2\| = \sqrt{4u_1^2 + u_2^2} = \sqrt{0+1} = 1.$$

$\therefore$ The orthonormal set is $\{(\frac{1}{2}, 0), (0, 1)\}$.

Co-ordinates Relative to Orthonormal Bases:

Theorem-6.3.1:

If $S = \{u_1, u_2, \dots, u_n\}$ is an orthonormal basis for an inner product space V , and u is any vector in V , then

(u is a linear combination of $(u_1, u_2, \dots, u_n)$)

$$u = \langle u, u_1 \rangle u_1 + \langle u, u_2 \rangle u_2 + \dots + \langle u, u_n \rangle u_n.$$

Then, the

coordinates of the vector u relative to orthonormal basis S is $(u)_S = (\langle u, u_1 \rangle, \langle u, u_2 \rangle, \dots, \langle u, u_n \rangle)$.

Eg-3: Co-ordinate Vector Relative to an Orthonormal Basis:

$$\text{Let } u_1 = (0, 1, 0) \quad ; \quad u_2 = \left(-\frac{4}{5}, 0, \frac{3}{5}\right) \quad ; \quad u_3 = \left(\frac{3}{5}, 0, \frac{4}{5}\right).$$

Check that $S = \{u_1, u_2, u_3\}$ is an orthonormal basis for $\mathbb{R}^3$ with the Euclidean inner product. Express the vector $u = (1, 1, 1)$ as a linear combination of the vectors in S , and find the coordinate vector $(u)_S$.

Soln

$$\langle u, u_1 \rangle = 1, \quad \langle u, u_2 \rangle = -\frac{1}{5}, \quad \langle u, u_3 \rangle = \frac{7}{5}.$$

$\therefore$ by Theorem-6.3.1, we have

$$u = u_1 - \frac{1}{5}u_2 + \frac{7}{5}u_3.$$

$$\text{i.e. } (1, 1, 1) = (0, 1, 0) - \frac{1}{5}\left(-\frac{4}{5}, 0, \frac{3}{5}\right) + \frac{7}{5}\left(\frac{3}{5}, 0, \frac{4}{5}\right)$$

$\therefore$ The coordinate vector of u relative to S is.

$$(u)_S = (\langle u, u_1 \rangle, \langle u, u_2 \rangle, \langle u, u_3 \rangle) = \left(1, -\frac{1}{5}, \frac{7}{5}\right).$$

Note: For c is usually, of equation vectors in

Theorem-

If S is inner pro and $(u)_S =$

(a) $\|u\| =$

(b) $d(u, v) =$

(c) $\langle u, v \rangle =$

Ex-4: Cal

If $\mathbb{R}^3$ of the

In $\mathbb{R}^n$ then w relative

The nor part (a)

$\|u\|$

Co-ordinates

If $S = \{$

IP space

the orthon

Thus ,

$\{ \}$ is
 $4u_1 + u_2$
 set by

For a nonorthonormal bases, it is usually necessary to solve a system of equations in order to express a vector in terms of the basis.

Prakash
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Theorem - 6.3.2:

If S is an orthonormal basis for an n -dimensional inner product space, and if $(u)_S = (u_1, u_2, \dots, u_n)$ and $(v)_S = (v_1, v_2, \dots, v_n)$ then:

- (a) $\|u\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$
- (b) $d(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$
- (c) $\langle u, v \rangle = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

Ex-4: Calculating Norms using Orthonormal Bases:

If $\mathbb{R}^3$ has the Euclidean inner product, then the norm of the vector $u = (1, 1, 1)$ is $\|u\| = \sqrt{1} = 1$.

In Ex-3: if $\mathbb{R}^3$ have the orthonormal basis S , then we know that the coordinate vector of u relative to S is

$$(u)_S = \left(1, -\frac{1}{5}, \frac{7}{5} \right).$$

The norm can also be calculated from this vector using part (a) of Theorem-6.3.2.

$$\|u\| = \sqrt{1^2 + \left(-\frac{1}{5}\right)^2 + \left(\frac{7}{5}\right)^2} = \sqrt{\frac{75}{25}} = \sqrt{3}.$$

Coordinates Relative to Orthogonal Bases:

If $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal basis for an IP space V , then normalizing each of these vectors yields the orthonormal basis $S' = \left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \dots, \frac{v_n}{\|v_n\|} \right\}.$

Thus, from theorem 6.3.1.

is for an
 in V , then

Then, the
 $i) = (\langle u, v_1 \rangle, \langle u, v_2 \rangle, \dots, \langle u, v_n \rangle)$
 ral Basis

$\left(\frac{4}{5} \right)$.
 parts for $\mathbb{R}^3$
 vector
 vectors in

$$\frac{7}{5} \left(\frac{3}{5}, 0, \frac{4}{5} \right)$$

$$u = \langle u, \frac{v_1}{\|v_1\|} \rangle \frac{v_1}{\|v_1\|} + \dots + \langle u, \frac{v_n}{\|v_n\|} \rangle \frac{v_n}{\|v_n\|}$$

$$\Rightarrow u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u, v_2 \rangle}{\|v_2\|^2} v_2 + \dots + \frac{\langle u, v_n \rangle}{\|v_n\|^2} v_n$$

u is linear combination of the vectors in orthogonal basis S .

Thm-6.3.3:

If $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.

Ex 10.5

$$u = \langle u, \frac{v_1}{\|v_1\|} \rangle \frac{v_1}{\|v_1\|} + \dots + \langle u, \frac{v_n}{\|v_n\|} \rangle \frac{v_n}{\|v_n\|}$$

$$\Rightarrow u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u, v_2 \rangle}{\|v_2\|^2} v_2 + \dots + \frac{\langle u, v_n \rangle}{\|v_n\|^2} v_n$$

u is linear combination of the vectors in orthogonal bases S .

Thm-6.3.3:

If $S = \{v_1, v_2, \dots, v_n\}$ is an orthogonal set of nonzero vectors in an inner product space, then S is linearly independent.

Ex-6.5:

In Ex-2, we showed that the vectors $v_1 = (0, 1, 0)$, $v_2 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$ and $v_3 = (\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})$ form an orthonormal set with respect to the Euclidean IP on $\mathbb{R}^3$. By Thm-6.3.3, these vectors form a linearly independent set, and since $\mathbb{R}^3$ is 3-D, $S = \{v_1, v_2, v_3\}$ is an orthonormal basis for $\mathbb{R}^3$ (by Thm-5.45).

Ex-6.3:

9) Verify that the vectors $v_1 = (\frac{-3}{5}, \frac{4}{5}, 0)$, $v_2 = (\frac{4}{5}, \frac{3}{5}, 0)$, $v_3 = (0, 0, 1)$ form an orthonormal basis for $\mathbb{R}^3$ with the Euclidean IP. Express the following as linear combination of v_1, v_2, v_3 .

- (a) $(1, -1, 2)$ (b) $(3, -7, 4)$ (c) $(\frac{1}{7}, \frac{-3}{7}, \frac{5}{7})$

Soln: First verify v_1, v_2, v_3 are orthogonal.

$$\text{i.e. } \langle v_1, v_2 \rangle = 0, \langle v_2, v_3 \rangle = \langle v_3, v_1 \rangle = 0$$

From Thm-6.3.3 $S = \{v_1, v_2, v_3\}$ is linearly independent

(23) no. we c

prove

$\Rightarrow S$

$\Rightarrow$ for

1.

is $\parallel$

$\Rightarrow S$

$u =$

(a) u

(b) $u =$

(10) Verify -

$v_3 = (1, 0, 0)$

for $\mathbb{R}^4$

as a

(a) $(1, 1, 1, 1)$

$u = \frac{v_1}{7} +$

(c) $(-\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

$u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 +$

(11) In each p given, Use to the

(a) $w = (3, 7, 4, 1)$

(b) $(w)_S =$

$$\frac{\langle u, u_n \rangle}{\|u_n\|^2}$$

$$\frac{\langle u, u_n \rangle}{\|u_n\|^2} u_n$$

is orthogonal

of nonzero
is linearly

$$, 0, \frac{1}{\sqrt{2}})$$

the
form a

D,
by Thm 5.4.5)

$$, \frac{3}{5}, 0)$$

with the

combination

(no. of vectors = dimension of the V.S

we can either verify if L.I or span

prove that it is a basis)

$\Rightarrow S$ is a basis for $\mathbb{R}^3$ with EIP.
orthogonal

$\Rightarrow$ for orthonormal, check whether norm of each vector is 1.

$$\text{is } \|v_1\|=1; \|v_2\|=1; \|v_3\|=1.$$

$\Rightarrow S$ is an orthonormal basis.

$$u = \langle u, v_1 \rangle v_1 + \langle u, v_2 \rangle v_2 + \langle u, v_3 \rangle v_3$$

$$\textcircled{a} \quad u = \frac{-7}{5} v_1 + \frac{1}{5} v_2 + 2 v_3$$

$$\textcircled{b} \quad u = \frac{-37}{5} v_1 - \frac{9}{5} v_2 + 4 v_3$$

$$\textcircled{c} \quad u = \frac{-3}{7} v_1 - \frac{1}{7} v_2 + \frac{5}{7} v_3$$

10) Verify that the vectors $v_1 = (1, -1, 2, -1)$, $v_2 = (-2, 2, 3, 2)$, $v_3 = (1, 2, 0, -1)$, $v_4 = (1, 0, 0, 1)$ form an orthogonal basis for $\mathbb{R}^4$, with the Euclidean IP, then express the following as a linear combination of v_1, v_2, v_3 and v_4 .

$$\textcircled{a} \quad (1, 1, 1, 1)$$

$$\textcircled{b} \quad (\sqrt{2}, -3\sqrt{2}, 5\sqrt{2}, -\sqrt{2})$$

$$u = \frac{v_1}{7} + \frac{5v_2}{21} + \frac{v_3}{3} + v_4$$

$$u = \frac{15\sqrt{2}}{7} v_1 + \frac{5\sqrt{2}}{21} v_2 - \frac{2\sqrt{2}}{3} v_3$$

$$\textcircled{c} \quad \left(\frac{-1}{3}, \frac{2}{3}, \frac{-1}{3}, \frac{4}{3} \right) \rightarrow u = \frac{-3}{7} v_1 + \frac{11}{63} v_2 - \frac{1}{18} v_3 + \frac{1}{2} v_4$$

$$\left[u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 + \dots + \frac{\langle u, v_n \rangle}{\|v_n\|^2} v_n \right]$$

11) In each part an orthonormal basis relative to the Euclidean IP is given. Use Thm (6.3.1) to find the coordinate vector of w , w.r.t to that basis.

$$\textcircled{a} \quad w = (3, 7); \quad u_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad u_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$(w)_S = (\langle w, u_1 \rangle, \langle w, u_2 \rangle) = (-2\sqrt{2}, 5\sqrt{2})$$

(b) $w = (-1, 0, 2)$ $u_1 = \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right)$; $u_2 = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$
 $u_3 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$

Sol. $(w)_S = (\langle w, u_1 \rangle, \langle w, u_2 \rangle, \langle w, u_3 \rangle) = (0, -2, 1)$

(12) Let $\mathbb{R}^2$ have the Euclidean IP, and let $S = \{w_1, w_2\}$ be the orthonormal basis with $w_1 = \left(\frac{3}{5}, -\frac{4}{5}\right)$, $w_2 = \left(\frac{4}{5}, \frac{3}{5}\right)$

(a) Find the vectors u and v that have co-ordinate vectors $(u)_S = (1, 1)$; $(v)_S = (-1, 4)$.

(b) Compute $\|u\|$, $d(u, v)$ & $\langle u, v \rangle$ by thm 6.3.2 ; then check the result by performing the computations directly on u & v .

Sol. (a) given $\langle u, w_1 \rangle = 1$ & $\langle u, w_2 \rangle = 1$.

$\therefore u = \langle u, w_1 \rangle w_1 + \langle u, w_2 \rangle w_2 = 1\left(\frac{3}{5}, -\frac{4}{5}\right) + 1\left(\frac{4}{5}, \frac{3}{5}\right)$
 $u = \left(\frac{7}{5}, -\frac{1}{5}\right)$

$\langle v, w_1 \rangle = -1$; $\langle v, w_2 \rangle = 4$.

$\therefore v = \langle v, w_1 \rangle w_1 + \langle v, w_2 \rangle w_2 = (-1)\left(\frac{3}{5}, -\frac{4}{5}\right) + 4\left(\frac{4}{5}, \frac{3}{5}\right)$
 $= \left(\frac{13}{5}, \frac{16}{5}\right)$.

(b)

$\|u\| = \sqrt{u_1^2 + u_2^2} = \sqrt{2}$;

$d(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2} = \sqrt{(1+1)^2 + (1-4)^2} = \sqrt{4+9}$
 $= \sqrt{13}$

$\langle u, v \rangle = u_1 v_1 + u_2 v_2$
 $= (1)(-1) + (1)(4) = 3$.

(13) Let $\mathbb{R}^3$ have the Euclidean inner product, and let $S = \{w_1, w_2, w_3\}$ be the orthonormal basis with $w_1 = \left(0, \frac{3}{5}, \frac{4}{5}\right)$, $w_2 = (1, 0, 0)$ and $w_3 = \left(0, \frac{4}{5}, \frac{3}{5}\right)$

(a) Find the vectors u, v and w that have the co-ordinate vectors $(u)_S = (-2, 1, 2)$; $(v)_S = (3, 0, -2)$ and $(w)_S = (5, -4, 1)$.

(b) Comp by appl co-ordin then che directly

Sol. (a)

$u =$
 $=$

$u =$

$v =$

(b) $\|u\|$

(14) In each four di informati
 (a) $(u)_S$

Sol. $\|u\|$

$\|v-w\|$

$\|v+w\|$

$\langle v, u \rangle$

(b) $(u)_S$

$\|u\|$

$\langle v, u \rangle$

(15) (a) S.T 1 and $v_4 =$

(b) Use of t

⑥ Compute $\|u\|$, $d(u, w)$ and $\langle w, u \rangle$

by applying Theorem-6.3.2 to the co-ordinate vectors $(u)_S$, $(v)_S$ and $(w)_S$

then check the results by performing the computations directly on u, v and w .

Sol: ⑥ given $\langle u, w_1 \rangle = -2$, $\langle u, w_2 \rangle = 1$; $\langle u, w_3 \rangle = 2$.

$$\text{or } u = \langle u, w_1 \rangle w_1 + \langle u, w_2 \rangle w_2 + \langle u, w_3 \rangle w_3$$

$$= -2 \left(0, -\frac{3}{5}, \frac{4}{5} \right) + 1 \left(1, 0, 0 \right) + 2 \left(0, \frac{4}{5}, \frac{3}{5} \right)$$

$$= \left(0 + 1 + 0, \frac{6}{5} + 0 + \frac{8}{5}, -\frac{8}{5} + 0 + \frac{6}{5} \right)$$

$$u = \left(1, \frac{14}{5}, -\frac{2}{5} \right)$$

$$v = \left(0, -\frac{17}{5}, \frac{6}{5} \right); w = \left(-4, -\frac{11}{5}, \frac{23}{5} \right)$$

$$\textcircled{b} \|u\| = \sqrt{13}; d(u, v) = 5\sqrt{3}, \langle w, v \rangle = 13.$$

⑭ In each part, S represents some orthonormal basis for a four dimensional inner product space. Use the given information to find $\|u\|$, $\|v-w\|$, $\|v+w\|$ and $\langle v, w \rangle$.

$$\textcircled{a} (u)_S = (-1, 2, 1, 3); (v)_S = (0, -3, 1, 5); (w)_S = (-2, -4, 3, 1)$$

$$\text{Sol: } \|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2 + u_4^2} = \sqrt{1+4+1+9} = \sqrt{15}$$

$$\|v-w\| = \|(2, 1, -2, 4)\| = \sqrt{4+1+4+16} = \sqrt{25} = 5$$

$$\|v+w\| = \|(-2, -7, 4, 6)\| = \sqrt{4+49+16+36} = \sqrt{105}$$

$$\langle v, w \rangle = v_1 w_1 + v_2 w_2 + v_3 w_3 + v_4 w_4$$

$$= 0 + 12 + 3 + 5 = 20$$

$$\textcircled{b} (u)_S = (0, 0, -1, -1); (v)_S = (5, 5, -2, -2); (w)_S = (0, -3, 0)$$

$$\|u\| = \sqrt{2}; \|v-w\| = \sqrt{34}; \|v+w\| = \sqrt{18}$$

$$\langle v, w \rangle = 21$$

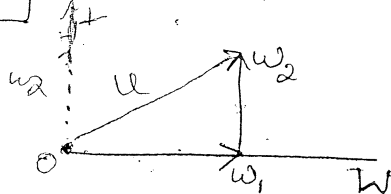
⑮ ① S.T the vectors $v_1 = (1, -2, 3, -4); v_2 = (2, 1, -4, -3); v_3 = (-3, 4, 1, -2)$ and $v_4 = (4, 3, 2, 1)$ form an orthonormal basis for $\mathbb{R}^4$ with the EIP.

② Use ① to express $u = (-1, 2, 3, 7)$ as a linear combination

Orthogonal Projections

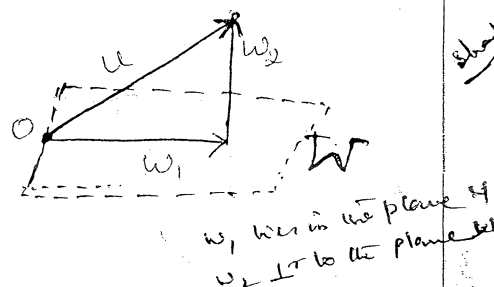
Thm - 6.3.4 : (Projection Thm) :

If W is a finite dimensional subspace of an IP space V , then every vector u in V can be expressed in exactly one way as $u = w_1 + w_2$ where w_1 is in W and w_2 is in $W^\perp$.



Note:

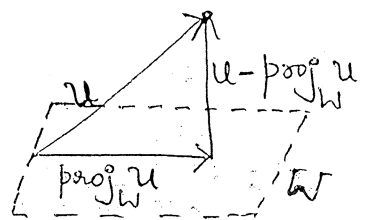
The vector w_1 in W is called orthogonal projection of u on W and it is denoted by $\text{proj}_W(u)$, and the vector w_2 in $W^\perp$ is called component of u orthogonal to W and it is denoted by $\text{proj}_{W^\perp}(u)$.



$$i.e. \quad u = \text{proj}_W(u) + \text{proj}_{W^\perp}(u).$$

$$\Rightarrow \text{proj}_{W^\perp}(u) = u - \text{proj}_W(u).$$

$$\Rightarrow u = \text{proj}_W(u) + (u - \text{proj}_W(u))$$



Thm - 6.3.5 :-

Let W be a finite dimensional subspace of an IP space V ,

① If $\{v_1, v_2, \dots, v_r\}$ is an orthonormal basis for W and u is any vector in V , then

$$\text{proj}_W u = \langle u, v_1 \rangle v_1 + \langle u, v_2 \rangle v_2 + \dots + \langle u, v_r \rangle v_r$$

② If $\{v_1, v_2, \dots, v_r\}$ is an orthogonal basis for W , and u is any vector in V , then

$$\text{proj}_W u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u, v_2 \rangle}{\|v_2\|^2} v_2 + \dots + \frac{\langle u, v_r \rangle}{\|v_r\|^2} v_r.$$

Do problems (21), (22), (23).

Thm: 6.

Every IP space

Proof:

Let V be a finite dimensional IP space. Let $\{u_i\}$ be a basis of V .

Step-1:

Step-2: u_1

$u_2 =$ where u_2 is orthogonal to u_1 and u_1, u_2 spanned W .

$\Rightarrow$

Step-3: u_1

is orthogonal to u_1, u_2 and u_1, u_2, u_3 spanned W .

Step-4: u_1

we find

where u_1 is a basis of W . Continuing set of

Thm: 6.3.6:

Every non-zero finite dimensional IP space has an orthonormal basis.

Proof: Gram-Schmidt process:

Let V be any non-zero finite dimensional IP space and let $\{u_1, u_2, \dots, u_n\}$ is any basis for V . The following sequence of steps will produce an orthogonal basis $\{v_1, v_2, \dots, v_n\}$ for V .

Step-1: Let $v_1 = u_1$.

Step-2: We can obtain v_2 orthogonal to v_1 by

$$v_2 = u_2 - \text{proj}_{W_1}(u_2) \quad (\text{Thm-6.3.5})$$

where v_2 is orthogonal to the subspace W_1 spanned by v_1 .

$$\Rightarrow v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$$

Step-3: We construct a vector v_3 which is orthogonal to both v_1 and v_2 and is given by,

$$v_3 = u_3 - \text{proj}_{W_2}(u_3) \\ = u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2 \quad (\text{see fig.})$$

where v_3 is orthogonal to the subspace W_2 spanned by v_1 and v_2 .

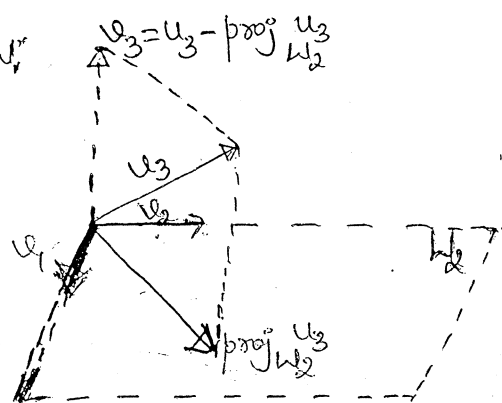
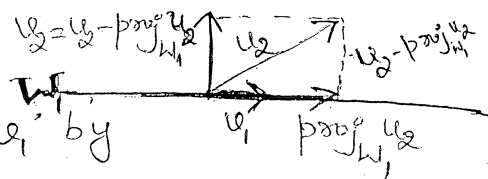
Step-4: To determine a vector v_4 which is orthogonal to v_1, v_2, v_3 ,

we find that by, $v_4 = u_4 - \text{proj}_{W_3}(u_4)$.

$$v_4 = u_4 - \frac{\langle u_4, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_4, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle u_4, v_3 \rangle}{\|v_3\|^2} v_3$$

where v_4 is $\perp$ to W_3 which is spanned by v_1, v_2, v_3 .

Continuing this process, we will obtain, after n steps, an orthogonal set of vectors $\{v_1, v_2, \dots, v_n\}$.



Since V is n -dimensional and every orthogonal set is linearly independent.

$\Rightarrow \{v_1, v_2, \dots, v_n\}$ is an orthogonal basis for V .
Orthonormal basis for V is obtained by

$$q_1 = \frac{v_1}{\|v_1\|}, q_2 = \frac{v_2}{\|v_2\|}, \dots, q_n = \frac{v_n}{\|v_n\|}.$$

$\Rightarrow \{q_1, q_2, \dots, q_n\}$ is an orthonormal basis of V .

Note: Advantage:

For $k \geq 2$, the following relationships are true.

1) $\{q_1, q_2, \dots, q_k\}$ is an orthonormal basis for the space spanned by $\{v_1, v_2, \dots, v_k\}$.

2) q_k is orthogonal to the space spanned by $\{v_1, v_2, \dots, v_{k-1}\}$.

Example-7: Using the Gram-Schmidt process.

Consider the vector space $\mathbb{R}^3$ with the Euclidean inner product. Apply the Gram-Schmidt process to transform the basis vectors $u_1 = (1, 1, 1)$; $u_2 = (0, 1, 1)$; $u_3 = (0, 0, 1)$ into an orthogonal basis $\{v_1, v_2, v_3\}$; then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{q_1, q_2, q_3\}$.

Soln:- Step-1: $v_1 = u_1 = (1, 1, 1)$

Step-2: $v_2 = u_2 - \text{proj}_{u_1} u_2$

$$= u_2 - \frac{\langle u_2, u_1 \rangle}{\|u_1\|^2} u_1$$

$$= (0, 1, 1) - \frac{2}{3} (1, 1, 1)$$

$$v_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

$$\langle u_2, u_1 \rangle = 0 + 1 + 1 = 2$$

$$\|u_1\|^2 = \|u_1\|^2 = 3.$$

(26)

v

$\Rightarrow v_2$

v_3

Thus

Ortho

The

$$\|v_1\|^2 =$$

$\therefore$ The

$$q_1 = \frac{v_1}{\|v_1\|}$$

$$q_3 =$$

Ex: 6.

(21) The

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Express

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onal set.

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basis of V.

True.

for the

y $\{u_1, u_2, \dots, u_{k-1}\}$

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ansform

$(0, 0, 1)$

normalize

orthonormal

Step-3: $v_3 = u_3 - \text{proj}_{u_1} u_3 - \text{proj}_{u_2} u_3$; $\|u_3\|^2 = \frac{4}{9} + \frac{1}{9} + \frac{1}{9} = \frac{6}{9} = \frac{2}{3}$

$$v_3 = u_3 - \frac{\langle u_3, u_1 \rangle}{\|u_1\|^2} u_1 - \frac{\langle u_3, u_2 \rangle}{\|u_2\|^2} u_2$$

$$\Rightarrow v_3 = (0, 0, 1) - \frac{1}{3} (1, 1, 1) - \frac{1/3}{2/3} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$= (0, 0, 1) - \frac{1}{3} (1, 1, 1) - \frac{1}{2} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$v_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

Thus $\left\{ v_1 = (1, 1, 1), v_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right), v_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right) \right\}$ is an

orthogonal basis for $\mathbb{R}^3$.

The norms of these vectors are;

$$\|u_1\| = \sqrt{3} ; \|u_2\| = \frac{\sqrt{6}}{3} \quad \|u_3\| = \frac{1}{\sqrt{2}}$$

$\therefore$ The orthonormal basis for $\mathbb{R}^3$ is

$$q_1 = \frac{u_1}{\|u_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) ; q_2 = \frac{u_2}{\|u_2\|} = \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$q_3 = \frac{u_3}{\|u_3\|} = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

Ex: 6.3

- (21) The subspace of $\mathbb{R}^3$ spanned by the vectors $u_1 = \left(\frac{4}{5}, 0, -\frac{3}{5}\right)$ and $u_2 = (0, 1, 0)$ is a plane passing through the origin. Express $w = (1, 2, 3)$ in the form $w = w_1 + w_2$ where w_1 lies in the plane and w_2 is perpendicular to the plane.

Sol:

$$\langle w, u_1 \rangle = 0$$

~~$\therefore$ The subspace $\mathbb{R}^3$ spanned by $\{u_1, u_2\}$ is an~~

~~$\therefore$ The set $\{u_1, u_2\}$ is an orthonormal basis set~~

$$w = w_1 + w_2 = \text{proj}_{\mathbb{R}^3} w + \text{proj}_{\mathbb{R}^3} w$$

$$0+1+1=2$$

$$\|w\|^2 = \sqrt{14}$$

Soln: $\langle u_1, u_2 \rangle = 0$ $\|u_1\| = 1$; $\|u_2\| = 1$ and forms a basis.
 $\therefore u_1, u_2$ are orthonormal vectors. Let Z be the subspace spanned by $\{u_1, u_2\}$.

$$W = w_1 + w_2 \\ = \text{proj}_Z W + (W - \text{proj}_Z W)$$

From Thm-6.3.5,

$$w_1 = \text{proj}_Z W = \langle w, u_1 \rangle u_1 + \langle w, u_2 \rangle u_2$$

$$= -1 \left(\frac{4}{5}, 0, -\frac{3}{5} \right) + 2(0, 1, 0) \\ = \left(\frac{4}{5}, 2, \frac{3}{5} \right)$$

$$w_2 = W - \text{proj}_Z(W) = (1, 2, 3) - \left(\frac{4}{5}, 2, \frac{3}{5} \right) \\ = \left(\frac{1}{5}, 0, \frac{12}{5} \right)$$

$$\therefore W = w_1 + w_2$$

② Repeat Exercise ① with $u_1 = (1, 1, 1)$ and $u_2 = (2, 0, -1)$

Here u_1, u_2 are orthogonal vectors and forms a basis.

$$W = w_1 + w_2 = \text{proj}_Z W + (W - \text{proj}_Z W)$$

From Thm-6.3.5,

$$w_1 = \text{proj}_Z W = \frac{\langle w, u_1 \rangle}{\|u_1\|^2} u_1 + \frac{\langle w, u_2 \rangle}{\|u_2\|^2} u_2$$

$$= \frac{2+1+1}{3} (1, 1, 1) + \frac{(-1)}{5} (2, 0, -1) \\ = (2, 2, 2) + \left(-\frac{2}{5}, 0, \frac{1}{5} \right) \\ = \left(\frac{8}{5}, 2, \frac{11}{5} \right)$$

$$w_2 = W - \text{proj}_Z W = (1, 2, 3) - \left(\frac{8}{5}, 2, \frac{11}{5} \right) \\ = \left(-\frac{3}{5}, 0, \frac{4}{5} \right)$$

$$\therefore W = w_1 + w_2$$

$$(1, 2, 3) - \left(\frac{9}{4}, 2, \frac{12}{8} \right) \\ = \left(-\frac{5}{4}, 0, \frac{7}{8} \right)$$

$$\langle w, u_1 \rangle = \frac{4}{5} + 0 - \frac{9}{5} = -1$$

$$\langle w, u_2 \rangle = 2$$

$$\langle u_1, u_2 \rangle = 2 + 0 - 1 = 1$$

$$\langle w, u_1 \rangle = 1 + 2 + 3 = 6$$

$$\|u_1\| = \sqrt{3}$$

$$\langle w, u_2 \rangle = 2 + 0 - 3 = -1$$

$$\|u_2\| = \sqrt{4+1} = \sqrt{5}$$

$$\frac{10-3}{5} = \frac{7}{5}$$

$$(1, 1, 1) (2, 0, -1)$$

$$2+0-2 = 0$$

23. Let 1

Express

$$w = w_1$$

w sp

w_2 is

Ex-6.3:

⑬ Let R^2 to three basis.

⑭ $u_1 =$

Soln &

Step-2

$\therefore \{u$
vectors

Since

linear

$\therefore \{$

Ortho

$q_i =$

$\therefore \{$

of

and a basis.

W be the
space spanned
by $\{u_1, u_2\}$

$$47 = \frac{4}{5} + 0 - \frac{9}{5} = -1$$

$$7 = 2$$

$$(2, 0, -1)$$

is a basis.

$$u_1, u_2 = 2 + 0 - 1 = 1$$

$$7 = 1 + 2 + 3 = 6$$

$$1 = \sqrt{3}$$

$$7 = 2 + 0 - 3 = -1$$

$$\|u_2\| = \sqrt{4+1} = \sqrt{5}$$

$$\frac{10-3}{5} = \frac{7}{5}$$

$$(2, 0, -1)$$

$$2+0-1$$

$$= 1$$

23. Let R^4 have the Euclidean IP.

Express $w = (-1, 2, 6, 0)$ in the form

$w = w_1 + w_2$, where w_1 is in the space

W spanned by $u_1 = (-1, 0, 1, 2)$; $u_2 = (0, 1, 0, 1)$ and

w_2 is orthogonal to W .

$$\langle u_1, u_2 \rangle =$$

Ex-6.3:

16. Let R^2 have the Euclidean IP. Use Gram-Schmidt process to transform the basis $\{u_1, u_2\}$ into an orthonormal basis.

(a) $u_1 = (1, -3)$; $u_2 = (2, 2)$ (b) $u_1 = (1, 0)$, $u_2 = (3, -5)$

Solve by Q. No Step-1: Let $u_1 = u_1 = (1, -3)$. $W_1 = \{u_1\}$

Step-2: $v_2 = u_2 - \text{proj}_{W_1} u_2$

$$= (2, 2) - \left\{ \frac{\langle u_2, u_1 \rangle}{\|u_1\|^2} u_1 \right\}$$

$$= (2, 2) - \left\{ \frac{-4}{10} (1, -3) \right\}$$

$$= (2, 2) + \left(\frac{4}{10}, \frac{-12}{10} \right) = \left(\frac{24}{10}, \frac{8}{10} \right)$$

$$\langle u_2, u_1 \rangle = \langle (2, 2), (1, -3) \rangle = 2 - 6 = -4$$

$$\|u_1\| = \sqrt{1+9} = \sqrt{10}$$

$$\left(\frac{12}{5} \right)^2 = \frac{144}{25}$$

$\therefore \{u_1, u_2\} = \left\{ (1, -3), \left(\frac{24}{10}, \frac{8}{10} \right) \right\}$ is an orthogonal set of vectors.

Since V is 2-dimensional and every orthogonal set is linearly independent.

$\therefore \{u_1, u_2\}$ is an orthogonal basis for R^2 .

Orthonormal basis for V is obtained by,

$$q_1 = \frac{u_1}{\|u_1\|} = \left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right) ; q_2 = \frac{u_2}{\|u_2\|} = \frac{\left(\frac{24}{10}, \frac{8}{10} \right)}{\frac{4}{5} \sqrt{10}}$$

$$= \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right)$$

$\therefore \left\{ \left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right), \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right) \right\}$ is an orthonormal basis of V .

17) Let $\mathbb{R}^3$ have Euclidean I.P. Use the Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into an orthonormal basis.

(a) $u_1 = (1, 1, 1)$, $u_2 = (-1, 1, 0)$; $u_3 = (1, 2, 1)$

Solve: Step-1: Let $v_1 = u_1 = (1, 1, 1)$ $W_1 = \{v_1\}$

Step-2: $v_2 = u_2 - \text{proj}_{W_1} u_2$

$\langle u_2, v_1 \rangle = -1 + 1 + 0 = 0$

$\|v_1\| = \sqrt{3}$

$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (-1, 1, 0) - 0$

$\Rightarrow v_2 = (-1, 1, 0)$ $\therefore W_2 = \{v_1, v_2\}$

Step-3: $v_3 = u_3 - \text{proj}_{W_2} u_3$

$= u_3 - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$

$\langle u_3, v_1 \rangle$

$= 1 + 2 + 1 = 4$

$\langle u_3, v_2 \rangle$

$= -1 + 2 + 0 = 1$

$\|v_2\| = \sqrt{2}$

$v_3 = (1, 2, 1) - \frac{4}{3}(1, 1, 1) - \frac{1}{2}(-1, 1, 0)$

$= \left(1 - \frac{4}{3} + \frac{1}{2}, 2 - \frac{4}{3} - \frac{1}{2}, 1 - \frac{4}{3} - 0\right)$

$v_3 = \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right)$

$\therefore$ The set $v_1 = (1, 1, 1)$, $v_2 = (-1, 1, 0)$, $v_3 = \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right)$ is an orthogonal set.

Since $\mathbb{R}^3$ is 3-D & every orthogonal set is L.I.

$\Rightarrow \{v_1, v_2, v_3\}$ is an orthogonal basis for $\mathbb{R}^3$.

$q_1 = \frac{v_1}{\|v_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$; $q_2 = \frac{1}{\sqrt{2}}(-1, 1, 0)$

$q_3 = \frac{v_3}{\|v_3\|} = \sqrt{6}\left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3}\right) = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$

$\therefore \{q_1, q_2, q_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) $u_1 = (1, 0, 0)$, $u_2 = (3, 7, -2)$; $u_3 = (0, 4, 1)$

Solve: $\left\{(1, 0, 0), \frac{1}{\sqrt{53}}(0, 2, 7), \frac{1}{\sqrt{53}}(0, 7, -2)\right\}$

18) Let $\mathbb{R}^4$ have product. L process to $\{u_1, u_2, u_3, u_4\}$ basis.

$u_1 = (3, 2, 1, 0)$

$u_3 = (1, 2, 0, -1)$

Solve:

Let $v_1 = u_1$

Step-2:

$v_2 = u_2 - \text{proj}_{W_1} u_2$

$= u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$

$= (1, -1, 0, 0) - 0$

$v_2 = (1, -1, 0, 0)$

$W_2 = \{v_1, v_2\}$

Step-3:

$v_3 = u_3 - \text{proj}_{W_2} u_3$

$= (1, 2, 0, -1) -$

Solve:

$= (1, 2, 0, -1) -$

$v_3 = \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right)$

Step-4:

$v_4 = u_4 - \text{proj}_{W_3} u_4$

$= (1, 0, 0, 1) - \frac{\langle u_4, v_1 \rangle}{\|v_1\|^2} v_1$

$= (1, 0, 0, 1) - 0 = (1, 0, 0, 1)$

$v_4 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\{v_1, v_2, v_3, v_4\}$

$q_1 = \frac{v_1}{\|v_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$q_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Gram-Schmidt

$$u_3 = (1, 2, 0)$$

$$-1 + 1 + 0 = 0$$

$$= \sqrt{2}$$

$$\langle u_3, v_1 \rangle$$

$$1 + 2 + 1 = 4$$

$$\langle u_3, v_2 \rangle$$

$$= -1 + 2 + 0 = 1$$

$$\|v_2\| = \sqrt{2}$$

$$\left(\frac{1}{6}, -\frac{1}{3} \right)$$

I.

V.

$$\|v_3\| = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{4}}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

18) Let $\mathbb{R}^4$ have the Euclidean inner product. Use the Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3, u_4\}$ into an orthonormal basis.

$$u_1 = (0, 2, 1, 0) \quad u_2 = (1, -1, 0, 0)$$

$$u_3 = (1, 2, 0, -1) \quad u_4 = (1, 0, 0, 1)$$

Soln:-

$$\text{Let } v_1 = u_1 = (0, 2, 1, 0)$$

$$w_1 = \{v_1\}; \|v_1\| = \sqrt{5}$$

Step-2:

$$v_2 = u_2 - \text{proj}_{w_1} u_2$$

$$= u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (1, -1, 0, 0) - \frac{(-2)}{5} (0, 2, 1, 0)$$

$$= (1, -1, 0, 0) - \frac{(-2)}{5} (0, 2, 1, 0)$$

$$v_2 = (1, -\frac{1}{5}, \frac{2}{5}, 0)$$

$$\|v_2\| = \sqrt{6/5}$$

$$w_2 = \{v_1, v_2\}$$

$$\text{Step-3: } v_3 = u_3 - \text{proj}_{w_2} u_3$$

$$= (1, 2, 0, -1) - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$= (1, 2, 0, -1) - \frac{4}{5} (0, 2, 1, 0) - \frac{3}{2} (1, -\frac{1}{5}, \frac{2}{5}, 0)$$

$$= (1, 2, 0, -1) - \frac{4}{5} (0, 2, 1, 0) - \frac{3}{2} (1, -\frac{1}{5}, \frac{2}{5}, 0)$$

$$v_3 = (\frac{1}{2}, \frac{1}{2}, -1, -1) \quad \|v_3\| = \sqrt{5/2}$$

Step-4:

$$v_4 = u_4 - \text{proj}_{w_3} u_4$$

$$= (1, 0, 0, 1) - \frac{\langle u_4, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_4, v_2 \rangle}{\|v_2\|^2} v_2 - \frac{\langle u_4, v_3 \rangle}{\|v_3\|^2} v_3$$

$$= (1, 0, 0, 1) - 0 - \frac{1}{6} (1, -\frac{1}{5}, \frac{2}{5}, 0) - 0$$

$$v_4 = (\frac{4}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{5}) \quad \|v_4\| = \sqrt{1/5}$$

$\therefore \{v_1, v_2, v_3, v_4\}$ is an orthonormal set

$$q_1 = \frac{v_1}{\|v_1\|} = (0, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0) \quad q_2 = (\frac{5}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{2}{\sqrt{30}}, 0)$$

$$q_3 = (\frac{1}{\sqrt{10}}, \frac{1}{\sqrt{10}}, \frac{-2}{\sqrt{10}}, \frac{-2}{\sqrt{10}}) \quad q_4 = (\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}) \quad w_3 = (\frac{1}{\sqrt{5}}, \frac{9}{\sqrt{5}}, \frac{19}{\sqrt{5}}, \frac{-9}{\sqrt{5}})$$

19) Let $\mathbb{R}^3$ have the Euclidean inner product. Find an orthonormal basis for the subspace spanned by $(0, 1, 2), (-1, 0, 1), (-1, 1, 3)$.

$$\text{Soln: } (0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}), (\frac{-\sqrt{5}}{\sqrt{6}}, \frac{-2}{\sqrt{30}}, \frac{1}{\sqrt{30}})$$

22) Repeat Ex 21 with $u_1 = (1, 1, 1)$ and $u_2 = (2, 0, -1)$

Soln:-

Here u_1, u_2 are not orthogonal. $\therefore$ First we find an orthogonal basis.

Step-1: Let $v_1 = u_1 = (1, 1, 1)$

$$\text{Step-2: } v_2 = u_2 - \text{proj}_{w_1} u_2$$

$$v_2 = (2, 0, -1) - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = (2, 0, -1) - \frac{1}{3} (1, 1, 1)$$

$$= (\frac{5}{3}, -\frac{1}{3}, -\frac{4}{3})$$

$\therefore \{v_1, v_2\}$ are orthogonal basis set of vectors.

$$w_1 = \frac{\langle w, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle w, v_2 \rangle}{\|v_2\|^2} v_2$$

$$= (\frac{13}{14}, \frac{31}{14}, \frac{40}{14})$$

$$w_2 = w_1 - \text{proj}_2 w_1 = (\frac{1}{14}, \frac{-3}{14}, \frac{2}{14})$$

23) Let $\mathbb{R}^4$ have the Euclidean IP.

Express $w = (-1, 2, 6, 0)$ in the form $w = w_1 + w_2$, where $w_1 \in W$ spanned by $u_1 = (-1, 0, 1, 2)$ and $u_2 = (0, 1, 0, 1)$ and w_2 is orthogonal to W .

$$\text{Soln: } \langle u_1, u_2 \rangle = 0 + 0 + 0 + 2 = 2 \neq 0$$

$$w_1 = (-\frac{5}{4}, \frac{1}{4}, \frac{5}{4}, \frac{9}{4})$$

Let R^3 have the IP $\langle u, v \rangle = u_1 v_1 + 2u_2 v_2 + 3u_3 v_3$. Use the Gram-Schmidt process to transform $u_1 = (1, 1, 1)$, $u_2 = (1, 1, 0)$, $u_3 = (1, 0, 0)$ into an orthonormal basis.

Soln: Step-1: Let $v_1 = u_1 = (1, 1, 1)$.
 $W = \{v_1\}$ $\|v_1\| = \sqrt{1+2+3} = \sqrt{6}$.
Step-2: $v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$
 $v_2 = (1, 1, 0) - \frac{\sqrt{6}}{3} (1, 1, 1)$ $\langle u_2, v_1 \rangle = 1+2+0 = 3$
 $= (1, 1, 0) - \frac{\sqrt{6}}{3} (1, 1, 1)$
Step-3: $v_3 = u_3 - \text{proj}_{W_2} u_3$ $W_2 = \{v_1, v_2\}$
 $v_3 = (1, 0, 0) - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$ $\langle u_3, v_1 \rangle = 1+0+0 = 1$
 $v_3 = (1, 0, 0) - \frac{1}{3} (1, 1, 1)$ $\langle u_3, v_2 \rangle = 0$
 $= (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})$

Soln: Step-1: Let $v_1 = u_1 = (1, 1, 1)$
 $W_1 = \{v_1\}$ and $\|v_1\| = \sqrt{u_1^2 + 2u_2^2 + 3u_3^2} = \sqrt{6}$.
Step-2:
 $v_2 = u_2 - \text{proj}_{W_1} u_2$
 $= (1, 1, 0) - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1$
 $= (1, 1, 0) - \frac{3}{6} (1, 1, 1)$
 $= (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ $\|v_2\| = 1$ $\langle u_3, v_1 \rangle = 1$
Step-3: $v_3 = u_3 - \text{proj}_{W_2} u_3$ $\langle u_3, v_2 \rangle = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}) \cdot (1, 0, 0) = \frac{1}{2}$
 $v_3 = (1, 0, 0) - \frac{\langle u_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle u_3, v_2 \rangle}{\|v_2\|^2} v_2$
 $= (1, 0, 0) - \frac{1}{6} (1, 1, 1) - \frac{1}{2} \cdot \frac{2}{\sqrt{3}} (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$

$$v_3 = (\frac{2}{3}, -\frac{1}{3}, 0)$$

$\therefore$ The orthogonal basis is $(1, 1, 1), (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}), (\frac{2}{3}, -\frac{1}{3}, 0)$

The orthonormal basis is
 $q_1 = \frac{1}{\sqrt{6}} (1, 1, 1); q_2 = \frac{1}{\sqrt{6}} (1, 1, -1)$
 $q_3 = \frac{1}{\sqrt{6}} (2, -1, 0)$

QR-DECOMPOSITION:

If 'A' is $m \times n$ matrix with L.I. column vectors, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix with orthonormal column vectors, and R is an $n \times n$ invertible upper triangular matrix.

ie, If $A = [u_1 | u_2 | \dots | u_n]$ $\leftarrow$ columns $m \times n$

$$Q = [q_1 | q_2 | \dots | q_n] \leftarrow \text{columns}$$

vectors in Q are orthonormal

$\therefore$ vectors in A (from Thm 6.3.1) can be expressed as,

$$u_1 = \langle u_1, q_1 \rangle q_1 + \langle u_1, q_2 \rangle q_2 + \dots + \langle u_1, q_n \rangle q_n$$

$$u_2 = \langle u_2, q_1 \rangle q_1 + \langle u_2, q_2 \rangle q_2 + \dots + \langle u_2, q_n \rangle q_n$$

$$\vdots$$

$$u_n = \langle u_n, q_1 \rangle q_1 + \langle u_n, q_2 \rangle q_2 + \dots + \langle u_n, q_n \rangle q_n$$

$$[u_1 | u_2 | \dots | u_n] = [q_1 | q_2 | \dots | q_n] \begin{bmatrix} \langle u_1, q_1 \rangle & \dots & \langle u_1, q_n \rangle \\ \langle u_2, q_1 \rangle & \dots & \langle u_2, q_n \rangle \\ \vdots & & \vdots \\ \langle u_n, q_1 \rangle & \dots & \langle u_n, q_n \rangle \end{bmatrix}$$

$$\Rightarrow A = QR$$

By GSP for $j \geq 2$ the vector q_j is orthogonal to $u_1, u_2, \dots, u_{j-1}$

$$\therefore R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_1, q_2 \rangle & \dots & \langle u_1, q_n \rangle \\ \langle u_2, q_1 \rangle & \langle u_2, q_2 \rangle & \dots & \langle u_2, q_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle u_n, q_1 \rangle & \langle u_n, q_2 \rangle & \dots & \langle u_n, q_n \rangle \end{bmatrix}$$

$$[\langle u_1, q_1 \rangle \quad \langle u_1, q_2 \rangle \quad \dots \quad \langle u_1, q_n \rangle]$$

$K \times O: Q \times K$
 3×3 matrix

Find the

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Soln: $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Applying Gram-Schmidt process on column vectors

$$q_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & & \\ 0 & & \\ 0 & & \\ 0 & & \end{bmatrix} = \begin{bmatrix} 3/\sqrt{3} & & \\ 0 & & \\ 0 & & \\ 0 & & \end{bmatrix}$$

The Q is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & & \\ 1/\sqrt{3} & & \\ 1/\sqrt{3} & & \end{bmatrix}$$

Importance:

Kindly use computing

Advanced nu

Ex-8: QR-Decomposition of a 3x3 matrix

Find the QR decomposition of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Soln:-

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Applying Gram-Schmidt process with subsequent normalisation to these column vectors yields the orthonormal vectors,

$$q_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, q_2 = \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, q_3 = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}$$

$$= \begin{bmatrix} 3/\sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

∴ The QR decomposition of A is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 3/\sqrt{3} & 2/\sqrt{3} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

Importance:

Widely used in algorithms for computing eigenvalues of large matrices.
→ Advanced numerical methods of L.F. Als

Ex-6.3:

25 - 28 → Excluded

24) Find QR-decomposition of the matrix, where

$$a) A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

$$b) A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$$

$$c) A = \begin{bmatrix} 1 & 1 \\ -2 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & 8/\sqrt{234} \\ -2/3 & 11/\sqrt{234} \\ 2/3 & 7/\sqrt{234} \end{bmatrix} \begin{bmatrix} 3 & 1/3 \\ 0 & \sqrt{26}/3 \end{bmatrix}$$

$$d) A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & 2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} & 1/\sqrt{3} \\ 0 & 0 & 4/\sqrt{6} \end{bmatrix}$$

$$e) A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & \sqrt{2}/2\sqrt{19} & -3/\sqrt{19} \\ 1/\sqrt{2} & -\sqrt{2}/2\sqrt{19} & 3/\sqrt{19} \\ 0 & 3\sqrt{2}/\sqrt{19} & 1/\sqrt{19} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 3/\sqrt{2} & \sqrt{2} \\ 0 & \sqrt{19}/\sqrt{2} & 3\sqrt{2}/\sqrt{19} \\ 0 & 0 & 1/\sqrt{19} \end{bmatrix}$$

$$f) A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

Columns are not L.I.